

Schema-Agnostic Rule Extraction from Industrial Standard Operating Procedures: A Multi-Agent LLM Pipeline and a Content-First Evaluation

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August 8, 2026

Abstract

Extracting operating rules from industrial Standard Operating Procedures (SOPs) is difficult because the information is scattered across prose, data tables, and access matrices, and because every facility documents its rules in its own vocabulary. The prevailing way to obtain high accuracy is to write an extraction prompt around one corpus’s schema (its field names, its rule taxonomy, its identifier convention) but a prompt built that way cannot be applied to a facility it was not written for without being rewritten by hand. Evaluation compounds the problem: exact-match metrics penalise an extraction for using a different name for the same quantity, confusing nomenclature with correctness.

This thesis asks what rule extraction costs when nothing about the target schema is supplied. We present a multi-agent pipeline, implemented in LangGraph, that assumes only that its input is an operating document. It segments each document and numbers every content line; parallel *scout* agents read the numbered chunks and emit records whose field names are copied from the document’s own labels; a single *inducer* call reconciles those labels into one corpus-level schema without ever seeing a document; deterministic assembly emits one row per rule; and parallel *audit* agents re-read each record against the lines it cites and drop what is not stated there. No stage is given a target schema, a rule taxonomy, or a worked example.

Extraction is scored by $F1_{\text{content}}$, a content-first metric that collapses each ground-truth row and each predicted record into a text blob, blends SBERT cosine similarity with a symmetric numeric agreement term, and pairs the two sides by Hungarian optimal assignment, so that identifier and field-name conventions never enter the score. The metric’s discriminating power is measured rather than assumed: predictions scored against a foreign corpus’s ground truth collapse to 0.01–0.10, corrupting every numeric value costs 0.20 F1, padding a record with numbers it did not extract is penalised rather than rewarded, and the reported operating point is not a local maximum on any of the four corpora.

The pipeline is evaluated on one development corpus and three previously unseen external corpora (biogas digestion, reverse-osmosis desalination, and sulfuric acid production) with five repeats each and no per-corpus configuration of any kind. It reaches $F1_{\text{content}} = 0.719 \pm 0.000$ on the development corpus and 0.715 ± 0.023 , 0.850 ± 0.019 , and 0.649 ± 0.049 on the three external corpora. Because the hosted endpoint is non-deterministic for long generations, every figure is reported as a mean and standard deviation over repeats rather than as a single run.

An ablation that removes the orchestration while holding the model, the prompt’s extraction discipline and the evaluation constant — one call per whole document, no chunking, no schema induction, no assembly, no audit — scores 0.733, 0.762, 0.793 and 0.607 on the same four corpora, winning on two and losing on two for a mean difference of 0.010, well inside run-to-run variation, at three to seven times fewer requests. The reported accuracy is therefore attributable to withholding the schema and to the extraction discipline expressed in the prompt rather than to the graph that runs it. What the graph supplies instead is auditability: every record cites the numbered line it was read from, where the single-prompt records cite nothing, so the grounding check reported here cannot be performed on them at all. The architecture is thus best characterised as trading request count for traceability, both of which are quantified.

Set against this, a grid search over ten prompting paradigms and two models, in which the prompt is given the development corpus’s schema, taxonomy, identifier convention, and worked examples, reaches 0.949 ± 0.020 on that corpus, and cannot be applied to the other three as a comparable condition without being rewritten by hand. Notably, the two systems recover the ground truth equally well (recall 0.907 and 0.909); the gap is entirely precision, because the two disagree about what constitutes one rule and any surplus record is counted as an error whether or not it is correct. The gap is therefore best read as the measured price of requiring no per-domain configuration, and the schema-specified condition as an instance of the corpus-specific adaptation this work seeks to avoid.

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1. INTRODUCTION

1.1 Context and Motivation

The field of Knowledge Extraction (KE) has undergone a radical paradigm shift, evolving from rigid, domain-specific architectures toward fluid, generative paradigms. Historically, the identification of pre-defined entities and relations within unstructured text was heavily reliant on fixed ontologies, which created significant bottlenecks when processing the vast volume and variety of modern digital data. With the widespread adoption of transformer architectures, manual heuristics have been progressively replaced by deep contextual representations. Consequently, a robust technical foundation has been provided for the conversion of raw digital signals and textual documentation into structured Knowledge Graphs (KGs).

Within modern industrial manufacturing facilities, KGs serve as the critical cognitive backbone for intelligent Digital Twins (DTs). However, high-quality KG construction represents a critical bottleneck for knowledge-driven digital twins, as traditional approaches such as manual ontology engineering or centralized natural language processing (NLP) pipelines fail to scale while maintaining semantic richness. To construct these KGs, operational knowledge must be systematically extracted from heterogeneous sources, primarily Standard Operating Procedures (SOPs). These documents encode essential parameters, including alarm thresholds, maintenance rules, and access policies. Theoretically, all operational logic required to synchronize a digital twin with its physical counterpart is encoded within these texts. In practice, however, the automated extraction of this knowledge is severely obstructed by multiple challenges.

First, a high degree of structural heterogeneity is exhibited by SOPs within a single facility. Critical information is frequently dispersed across fundamentally different layouts; for instance, modern industrial facilities generate knowledge simultaneously in natural language documents, numerical sensor streams, and manufacturer datasheets. The effective processing of these distinct formats cannot be consistently achieved by a single, monolithic extraction prompt, as the information is spatially and semantically structured in fundamentally divergent ways.

Second, the evaluation of automated extraction in industrial contexts is highly complex. Throughout this thesis, “semantic aliasing” denotes the phenomenon whereby one and the same operational constraint appears under different surface forms in different places: the document states the rule in prose or in a table cell, the annotator records it under an identifier and field vocabulary of their own convention, and an extractor may emit it under yet another name, so that several renderings of a single rule share no exact string. In ground-truth annotations, rule identifiers, such as RULE-ST01-01 or MAINT-01, are often established as annotator conventions that rarely appear verbatim in the source text. When evaluation relies on exact-match F1 over identifiers, taxonomic nomenclature is confused with true semantic accuracy. Consequently, an “evaluation blind spot” is created, which risks penalizing operationally valid extractions while simultaneously failing to detect hallucinated but syntactically correct numeric thresholds.

Large Language Models (LLMs) are considered highly attractive for this extraction task, given their capacity to interpret complex natural-language instructions and adapt to new extraction patterns. Nonetheless, significant reliability issues persist; LLMs are prone to hallucinating numeric values absent from the source text and possess a “black-box” nature that obscures their decision-making processes. To mitigate the lim-

itations of monolithic generative models, recent research proposes collaborative multi-agent architectures, in which specialized software agents — document parsers, entity extractors, ontology alignment agents — are orchestrated to extract, validate, and integrate knowledge across heterogeneous formats. Recent literature highlights several successful implementations of this paradigm; for example, the KARMA framework deploys nine collaborative LLM agents for entity discovery and conflict resolution, while Tag2Graph is explicitly designed to mine KGs from industrial time-series data and tag definitions.

Bridging the gap between the generative flexibility of LLMs and the deterministic rigor required by industrial SOPs remains a primary experimental frontier. This thesis investigates the design, implementation and evaluation of a collaborative multi-agent system for that task, on four corpora from unrelated industrial domains: a nine-station packaging facility used for development, and three previously unseen corpora covering biogas digestion, reverse-osmosis desalination and sulfuric acid production.

1.2 Research Gaps

Although multi-agent systems have significantly expanded the capacity for generative information extraction, systematic evaluation of these pipelines is constrained by a lack of specialised benchmarks. The review in Section 3 indicates an absence of systematic architectures for agent specialisation in extraction from industrial documentation, and a lack of comparative studies on agent collaboration strategies. Most consequentially for this work, the reported accuracy of such pipelines is almost always obtained on the same corpus their prompts were written around, so it is not known what they achieve on documentation they were not designed for.

From an evaluation perspective, existing work fails to simultaneously address the unique demands of industrial SOP extraction. Specifically, three critical requirements remain unfulfilled by the current state of the art:

- **Extraction that is not written around one corpus.** Reported accuracy on rule extraction is typically obtained by encoding the evaluation corpus’s own schema, rule taxonomy and identifier convention into the prompt. Such a system is not transferable: applying it to a second facility requires a human to read that facility’s documents and rewrite the prompt. What a pipeline achieves when nothing about the target schema is supplied is rarely measured, and is the question this thesis addresses.
- **Evaluation across corpora rather than within one.** A single annotated corpus cannot distinguish a general extraction capability from a system tuned to the documents it is scored on. Establishing the distinction requires previously unseen corpora from unrelated domains, scored without any per-corpus adjustment, and repeated often enough to separate a real difference from endpoint noise.
- **A content-first metric whose own validity is demonstrated.** Scoring operational correctness independently of annotator-defined naming requires semantic matching (Hungarian assignment over sentence-embedding similarity), but such a metric can assign a flattering score without discriminating anything, because embeddings place any two documents of the same genre at a non-trivial similarity. A metric introduced for this purpose must therefore be accompanied by evidence that it collapses when the extraction is wrong, and by a statement of what it does not measure.

This thesis responds to these gaps with a pipeline that receives no schema, an evaluation spanning four corpora from unrelated industrial domains, and a metric whose discriminating power and blind spots are both measured and reported.

1.3 Contributions and Objectives

To address these limitations, this work makes four primary contributions:

- **Schema-Agnostic Multi-Agent Pipeline:** A LangGraph extraction pipeline whose only assumption about its input is that the input is an operating document. Field names are copied from each document’s own labels by parallel scout agents, reconciled into a corpus-level schema by a single inducer call that never sees a document, assembled deterministically, and checked by parallel audit agents against the numbered lines each record cites. No stage receives a target schema, a rule taxonomy, or a worked example, so the same pipeline invocation runs unchanged on a facility it has never seen.
- **Content-First Evaluation Metric:** $F1_{\text{content}}$, which collapses each ground-truth row and predicted record into a text blob, blends SBERT cosine similarity [20] with a symmetric numeric agreement term, and pairs the two sides by Hungarian optimal assignment. Neither side’s field names or identifier conventions enter the score, which is what makes a comparison possible at all when the extractor discovers its own vocabulary.
- **Empirical Validation of the Metric Itself:** A metric built on sentence embeddings can post a respectable score without discriminating anything, so its validity is measured rather than argued. Predictions scored against a foreign corpus’s ground truth fall to 0.01–0.04; corrupting every numeric value costs 0.20 F1; padding a record with numbers it did not extract lowers its score rather than raising it; and the reported threshold is not a local maximum on any of the four corpora, so it cannot have been chosen to flatter the system. The same study reports, with equal prominence, what the metric does *not* measure: word order within a record, which value is bound to which record, which slot a value occupies inside a record, and the correctness of the discovered category label.
- **Cross-Corpus Evaluation and the Cost of Schema-Agnosticism:** Evaluation on one development corpus and three previously unseen external corpora (biogas digestion, reverse-osmosis desalination, sulfuric acid production), five repeats each, with no per-corpus configuration. Against this, a grid search over ten prompting paradigms and two models whose prompt is given the development corpus’s schema, taxonomy, identifier convention and worked examples scores markedly higher on that corpus and is inapplicable to the others without being rewritten. Because both systems recover the ground truth to within 0.002 recall and differ chiefly in how finely they itemise it, the gap quantifies what requiring no per-domain configuration costs, and identifies the schema-specified condition as corpus-specific adaptation rather than a fair baseline.
- **An Ablation Isolating the Architecture, and the Trade-Off It Reveals:** a single schema-free prompt, one call per document, holding the model and the prompt’s extraction discipline constant

and removing only the orchestration. It matches the pipeline to within 0.010 mean F1 across the four corpora — winning on two, losing on two — at three to seven times fewer requests, establishing that the model performs schema-agnostic extraction unaided. What the orchestration supplies is not accuracy but auditability: line-level provenance for every record, without which the grounding check of Section 10.5 cannot be run, and a reconciled field vocabulary. Both are properties the metric prices at zero.

2. THEORETICAL BACKGROUND

In this chapter, the foundational concepts and computational tools upon which the remainder of the thesis is built are introduced. The theoretical framework encompasses several interconnected domains, including deep learning, natural language processing (NLP), agentic artificial intelligence, and Knowledge Graphs (KGs). The primary objective is to provide sufficient theoretical context to elucidate the design decisions detailed in subsequent chapters, ensuring comprehensibility without assuming prior expertise in any specific technological paradigm.

2.1 Large Language Models

Large Language Models (LLMs) are characterized as advanced deep learning models that have been extensively pre-trained on massive textual corpora to understand context and generate human-like responses. The foundational architecture of nearly all modern LLMs is the transformer, which utilizes a self-attention mechanism to process sequences more efficiently and accurately by relating any token in an input to any other token. Various architectures have been developed within this paradigm, including encoder-only models (e.g., BERT) for deep contextual understanding, decoder-only models (e.g., GPT) for text generation, and encoder-decoder models (e.g., T5) for highly flexible text-to-text transformations.

Two properties of particular relevance to this thesis are zero-shot and few-shot in-context learning [14]. In zero-shot settings, a model follows task instructions expressed entirely in natural language without any worked examples, enabling novel extraction tasks with no labeled data. Few-shot in-context learning extends this by embedding a small number of worked examples directly in the prompt, guiding the model toward the desired output format or extraction pattern without task-specific fine-tuning or weight updates. Both capabilities render LLMs attractive for knowledge extraction in industrial settings where annotated training corpora are scarce, a direct constraint here, where the four evaluation corpora provide between 26 and 86 annotated rules each, 229 in total. Chain-of-Thought prompting [22], which requires the model to produce explicit intermediate reasoning steps before its final answer, is a complementary technique that improves extraction reliability by making intermediate decisions inspectable. Together, zero-shot prompting, few-shot in-context learning, and Chain-of-Thought reasoning represent application-level enhancements that improve LLM utility without modifying model weights [2], the paradigm this thesis operates within entirely, as no fine-tuning or weight adaptation of any model is performed.

Despite these capabilities, LLMs exhibit significant limitations that are especially consequential in industrial contexts. The most critical is the generation of *hallucinations* [17]. Two distinct hallucination

modes are distinguished in the literature: *intrinsic hallucination*, where the output directly contradicts information present in the source document, and *extrinsic hallucination*, where the output introduces content that cannot be verified against the source at all. Both are dangerous for industrial SOP extraction, but extrinsic hallucination is particularly insidious: a numeric threshold value invented by the model and absent from the source text will not be flagged by any input-grounding check, yet could cause a safety alarm to fire at an incorrect value or fail to trigger when it should. A second limitation is difficulty with precise numerical reasoning: numeric values embedded in multi-column threshold tables with engineering units are susceptible to transcription errors in which adjacent row values are transposed or incorrect unit suffixes are appended. Third, the “black-box” nature of LLM inference makes it difficult to trace why a specific prediction was produced, complicating systematic debugging of extraction errors. The mitigation of these limitations (hallucination control, numeric grounding, and reasoning transparency) are primary design concerns addressed by the multi-agent pipeline proposed in Section 8.

Models are served through an OpenAI-compatible endpoint (Section 6.3.1); nothing in the pipeline depends on that endpoint being remote rather than local, and the same code runs against either. The choice carries one methodological consequence that shapes every reported figure: long generations are not reproducible under a fixed seed, so no result in this thesis is presented as a single number. Section 6.3.3 gives the mechanism and the consequences that follow from it.

2.2 Knowledge Graphs and Graph Databases

A Knowledge Graph (KG) is defined as a structured representation of knowledge where entities and their relations are linked within a graph. The core structural elements include nodes, which represent entities or concepts, and edges, which denote the relationships between these entities. Properties or attributes are frequently added to provide supplementary factual data, such as dates, locations, or numerical values. KGs provide a semantic framework that supports complex data queries and reasoning, successfully narrowing queries to small data subsets and maintaining high performance regardless of dataset growth. A central design philosophy is to connect *concepts rather than strings* [8], enabling semantic relationship traversal rather than keyword lookup; practical applications of this paradigm include search optimisation, question-answering systems, and recommendation engines. The overall quality and utility of a KG are strictly dependent on the quality of the data fed into it, a principle that directly motivates the content-first evaluation metric introduced in this thesis.

Constructing a KG from unstructured text relies on three core NLP tasks [3]. *Entity extraction* (Named Entity Recognition, NER) identifies key entities in a document (persons, organizations, locations, sensor identifiers, operational parameters) that become the nodes of the graph. *Relation extraction* then infers typed relationships between pairs of identified entities from their textual context, producing the labelled edges that encode operational constraints and causal dependencies. A third task, *entity resolution*, determines whether multiple text mentions refer to the same real-world entity, consolidating duplicate candidate nodes before they are committed to the graph. Whereas early methods relied on hand-crafted syntactic rules and sequence-labelling models such as Conditional Random Fields, modern approaches adapt pre-trained language models to each extraction task through supervised fine-tuning or, as in this thesis, zero- and few-

shot prompting. Two broad construction strategies are distinguished in practice [8]: an *NLP-only* approach that extracts subject–verb–object triples directly from text (maximally flexible but susceptible to entity disambiguation errors when the same surface form refers to different entities), and an *NLP-lite* approach that identifies named entities and resolves them against a structured knowledge base such as Wikidata to standardise relationship types (built-in disambiguation, but constrained to pre-specified relation types).

The organization of KG content is heavily influenced by Description Logics, utilizing a bifurcated structure:

- The **TBox** (terminological box) establishes the domain ontology, defining the schema, entity types (e.g., Component, Process, Parameter, Sensor), relationship types (e.g., *part_of*, *monitors*, *triggers*), and property constraints.
- The **ABox** (assertional box) contains the concrete instance data (the specific sensors, stations and extracted rules of one facility) populated against the schema the TBox defines.

Graph databases such as Neo4j implement this model natively and expose it through query languages such as Cypher, supporting the multi-hop traversal that downstream analytics require. That downstream stage is the motivation for this work rather than part of it: this thesis addresses the step that must succeed first (extracting the rules) and evaluates that step against annotated ground truth. Whether the extracted rules are operationally sufficient to drive graph-based analytics is a separate question, and Section 10.13 states plainly that it is not answered here.

A growing body of research explores the interaction between LLMs and KGs more broadly [19]. Three principal integration paradigms are identified: *LLM-augmented KGs*, where LLMs serve as the primary extraction engine to populate a graph from unstructured text; *KG-augmented LLMs*, where structured graph knowledge is injected into the LLM context to reduce hallucination and improve factual grounding; and *synergized frameworks*, where both modalities reinforce each other iteratively. This thesis sits within the first paradigm (LLMs as the extraction engine that would populate such a graph) and stops at the extraction boundary, where the output is a set of typed rule records with per-record provenance rather than a populated graph.

2.3 Multi-Agent Systems and LangGraph

To overcome the limitations of centralized NLP pipelines, which frequently fail to scale while maintaining semantic richness, a collaborative multi-agent paradigm is increasingly adopted. A multi-agent system (MAS) is defined as a software architecture comprising multiple autonomous, specialized components, termed agents, that interact to achieve complex objectives. Various collaboration strategies can be supported by these architectures, including cooperative, competitive, and hierarchical frameworks.

The primary advantage of a multi-agent approach is agent specialization. Within an extraction pipeline, dedicated agents can be deployed for specific tasks, such as document parsing, domain-specific named entity recognition (NER), relationship extraction, ontology alignment, and validation. Recent literature highlights the efficacy of this approach in industrial contexts. The KARMA framework [6] employs nine collaborative

LLM agents to mine a multi-source industrial dataset, extracting 38,230 entities at 83.1% LLM-verified correctness while achieving an 18.6% reduction in inter-agent entity conflicts through structured majority voting. Analogous frameworks such as Tag2Graph apply a collaborative paradigm to mine KGs directly from industrial time-series data and tag definitions. Multi-agent architectures have also demonstrated superior performance on standard NER and relation extraction benchmarks relative to single-agent baselines [10]. To ensure semantic consistency more broadly, conflict resolution mechanisms frequently utilize majority voting protocols with configurable confidence thresholds to assign composite reliability scores to extracted triples.

The extraction pipeline presented in this thesis is implemented using the LangGraph framework. LangGraph facilitates the organization of agents as a typed directed acyclic graph (DAG) with a shared state dictionary. Two features are load-bearing in this implementation. Parallel fan-out lets one agent role be instantiated once per document chunk and run concurrently, which is how both the extraction and the audit stages work. Sum-reducer state fields let those concurrent instances accumulate into a shared list without overwriting one another, which is what makes the fan-out safe: each scout appends its records rather than replacing the state.

2.4 Industrial Digital Twins and Standard Operating Procedures

A Digital Twin is conceptualized as a highly synchronized software model of a physical system, fundamentally requiring sophisticated knowledge graphs to integrate diverse static and dynamic data sources. The architecture of an agentic Digital Twin typically integrates a static model, represented by the knowledge graph defining the industrial system structure, with a dynamic model processing simulated IoT data and operational states. KGs serve as the cognitive backbone of these twins, enabling bidirectional synchronization, incremental updates, and complex what-if reasoning. Sahlab et al. [9] demonstrate that KG integration delivers four concrete capabilities for industrial Digital Twins: internal cross-document linking that connects entities across heterogeneous data sources, autonomous knowledge completion that infers missing relationships from existing graph structure, automated error detection in sensor readings by reasoning over expected operational states, and semantic querying of the facility model using ontology-aware graph traversal. Recently, the concept of the “Digital Twin Agent” has emerged, incorporating autonomous decision-making and cross-system collaboration to accelerate modeling and resilience in manufacturing environments.

Within industrial facilities, Standard Operating Procedures (SOPs) function as the primary governance layer, encoding the operational rules that dictate Digital Twin behavior. These natural language documents specify critical parameters, including continuous numerical measurement thresholds, maintenance rules, and zone access policies. However, the automated extraction of this logic is complicated by the extreme structural heterogeneity of SOPs, which frequently combine narrative prose, complex alarm threshold tables, and personnel access matrices. The translation of these heterogeneous, unstructured documents into verified, machine-readable rule records (the form a graph would subsequently be populated from) is the central methodological challenge addressed by the multi-agent system developed in this thesis.

3. RELATED WORK

This chapter reviews existing work in the three areas most relevant to this thesis: document-level information extraction and evaluation, Knowledge Graph construction from text, and Knowledge Graphs for industrial Digital Twins.

3.1 Document-Level Information Extraction and Evaluation

The most widely used benchmarks for document-level relation extraction are DocRED [23] and DocILE [21]. DocRED contains approximately 100,000 relational facts extracted from Wikipedia articles, with cross-sentence dependencies that require multi-paragraph reasoning. While DocRED pioneered large-scale joint entity and relation extraction datasets, it primarily targets encyclopedic relations without engineering analogs, lacking the logical depth required for industrial constraints. Conversely, DocILE provides the largest public benchmark for Key Information Localization and Extraction (KILE) and Line Item Recognition (LIR) across over 1,000 unique industrial layouts. Both have been important for advancing general-purpose information extraction systems, but neither addresses the characteristics specific to industrial SOP corpora: numeric sensor thresholds with engineering units, conditional operational logic distributed across heterogeneous document formats (prose, tables, matrices), and a need to evaluate extraction quality independently of how identifiers are named. Specifically, DocILE focuses on static business entities (e.g., tax totals, addresses) rather than the conditional operational rules and numeric sensor thresholds required for industrial SOP automation.

In the domain of business rules, BREX [1] represents the state-of-the-art for large-scale rule-flow modelling with conditional logic across 30+ domains. However, BREX ignores tabular structures and numeric engineering units. Furthermore, it relies on identifier-based F1 evaluation, which assumes that rule names in the source text match the ground-truth annotation convention. For industrial SOPs, rule identifiers are typically annotator-assigned rather than present in the source text, so identifier-based F1 rewards naming convention alignment as much as extraction quality. This is the core motivation for the $F1_{\text{content}}$ metric defined in Section 6.5, and for reporting identifier agreement only as a diagnostic of the annotation convention (Section 6.5.9) rather than as a score. For regulatory domains, De Jure [4] provides a robust framework for deontic logic (obligations and permissions), but it does not account for the physical threshold semantics essential for SCADA-level integration.

SOP-Bench [12] specifically targets industrial procedures and evaluates whether an agent can correctly *execute* a procedure step-by-step. This is a complementary task to the one addressed here: SOP-Bench measures task completion via action sequences, whereas this thesis addresses the prior data-fidelity step of extracting and structuring the operational rules that an execution agent would need to follow. Table 1 positions the evaluation setting used in this thesis relative to these benchmarks. The relevant contrast is not corpus size (the four corpora used here are small) but which properties of industrial rule extraction a benchmark exercises at all.

Table 1: Positioning of this thesis’s evaluation setting relative to related benchmarks. $\checkmark$ = fully addressed; $\circ$ = partially addressed; $\times$ = not addressed. *Schema-free* asks whether a system is evaluated without being given the target schema; no prior benchmark tests this, which is the gap this thesis addresses.

Benchmark	Large-scale	Primary Focus	Numeric Thresholds	Struct. Heterog.	Content-first Metric	Schem
DocRED [23]	$\checkmark$	Entity Relations	$\times$	$\times$	$\times$	$\times$
BREX [1]	$\checkmark$	Business Rules	$\times$	$\times$	$\times$	$\times$
DocILE [21]	$\checkmark$	Key-Value (KILE)	$\times$	$\checkmark$	$\times$	$\times$
De Jure [4]	$\checkmark$	Legal Compliance	$\times$	$\circ$	$\times$	$\times$
SOP-Bench [12]	$\times$	Agentic Execution	$\times$	$\checkmark$	$\times$	$\times$
This thesis	$\times$	Rule Ex-traction	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

3.2 LLM-Based Knowledge Graph Construction

The use of LLMs to extract graph-structured knowledge from unstructured text has received significant attention, representing a fundamental paradigm shift from early rule-based systems and probabilistic sequence labelling toward generative extraction enabled by deep bidirectional transformers and attention mechanisms. Several recent surveys identify multiple interaction patterns between LLMs and KGs, including LLMs used to populate KGs, KGs used to ground and verify LLM outputs, and joint architectures. This thesis falls in the first category. It stops at the extraction boundary: the output is a set of rule records carrying per-record provenance, which is the input a graph-population stage would consume, and the evaluation therefore scores those records against annotated ground truth rather than scoring a graph.

Traditional KG construction systems rely on fine-tuned Named Entity Recognition (NER) and Relation Extraction (RE) models trained on labelled corpora. These approaches require substantial annotated data, which none of the corpora used here provides: their ground truths hold between 26 and 86 rules, far too few to fine-tune a dedicated extraction model. The constraint is not incidental to this study but typical of the setting: a facility documenting its own procedures does not also produce thousands of labelled extractions. The prompt-engineering approach used here avoids this requirement entirely, leveraging emergent reasoning capabilities to convert raw digital signals into structured relational assets. This allows large language models to function as zero-shot learners or reasoning agents via Chain-of-Thought and ReAct paradigms, even in low-resource scenarios. The trade-off is that prompt-based LLMs are more sensitive to the exact phrasing of instructions than a trained model would be, which is part of the reason why the grid search over prompting paradigms (Section 7) was necessary. Furthermore, applying knowledge extraction to industrial SOPs remains challenging because manufacturing documents often embed critical data in non-narrative for-

mats. To overcome the structural heterogeneity of these documents, recent research has pivoted towards multi-agent architectures that use specialized agents to coordinate extraction across different formats. For example, TabAgent [13] proposes a multi-agent framework specifically for unstructured table extraction, while SOPRAG [11] utilizes multi-view graph experts to navigate complex SOP hierarchies. Other recent frameworks, such as SPIRES [15], employ recursive extraction methods that enforce strict schema adherence.

Hallucination control is a known challenge in LLM-based extraction that has received relatively little attention in the literature for numeric fields specifically. Populating knowledge graphs from unstructured technical text remains highly error-prone due to "hallucinations" in natural language generation. The cost of such errors is domain-dependent, and in industrial settings it is not tolerable: a hallucinated temperature threshold could cause a safety alarm to fire at the wrong value or not at all, so an invented numeric is a safety defect rather than a noise term. The grounding discipline used in this thesis (Section 8.5) addresses the most dangerous hallucination patterns for this task: inventing numeric threshold values, appending units not present in the source, and borrowing values from neighbouring rows, columns, or sentences. Rather than relying on a prompt instruction alone, every extracted record cites the numbered source line it was read from, and a separate audit agent re-reads the record against that line. The grounding instruction carried by every extraction prompt is deliberately compact: each numeric value and unit must be copied verbatim from the source text, and a field must be left empty rather than guessed. Compactness is itself the design choice: an instruction elaborate enough to enumerate the shapes a constraint may take teaches a model the phrasing of the corpus it was written for, which is the failure mode this pipeline exists to avoid.

3.3 Knowledge Graphs for Industrial Digital Twins

The integration of Knowledge Graphs with industrial Digital Twins has been studied in the context of fault detection, predictive maintenance, and process monitoring. Knowledge Graphs serve as the essential semantic backbone for Intelligent Digital Twins by structurally linking scattered technical data. A recurring pattern in this literature is the use of an ontology-based KG to link sensor readings to the operational rules that govern them, enabling graph-based queries such as “which rule covers this anomaly event?”.

A common limitation in this body of work is that the KG content (the operational rules, threshold values, and causal relationships) is populated manually by domain experts or from structured databases. Automating this step from unstructured SOP documents, which is the contribution of this thesis, has not been directly addressed. While recent literature demonstrates that agentic workflows are superior for automating SOP execution, a systematic evaluation of these extraction pipelines is hindered by a lack of specialized benchmarks. Current evaluation infrastructures, such as KGpipe [7] or Plumber [16], offer modular extraction pipelines but notably lack the manufacturing-specific ground truth necessary to validate SCADA-level constraints.

4. SYSTEM OVERVIEW

Before describing each component in detail, this section gives a high-level picture of the full pipeline so that the later sections are easier to follow.

The system takes a directory of operating documents as input and produces one row per extracted rule as output, together with the provenance of every row: the document and the numbered line each record was read from. The processing is organised in three layers, each described in a dedicated section:

Layer 1: Document Parsing (Section 5) Source PDFs are converted into clean plain-text files using IBM’s docling library for layout-aware extraction, followed by normalisation (removing OCR artefacts, fixing encoding errors), so that table structure and heading hierarchy survive into the text the extraction layer reads.

Layer 2a: Schema-Specified Grid Search (Section 7) A grid over ten prompting paradigms and two models in which the prompt is given the development corpus’s schema, rule taxonomy, identifier convention and worked examples. This establishes what is achievable when the target schema is known, and, because such a prompt cannot be applied to any other corpus without being rewritten by hand, it also delimits precisely what that accuracy is contingent upon.

Layer 2b: Multi-Agent Extraction Pipeline (Section 8) A LangGraph pipeline that receives no schema. Documents are segmented and every content line numbered; parallel scout agents emit records whose field names are copied from each document’s own labels; one inducer call reconciles those labels into a corpus-level schema without seeing any document; assembly is deterministic; and parallel audit agents re-read each record against the lines it cites.

Layer 3: Evaluation (Section 6) Extracted rules are scored against each corpus’s ground truth by $F1_{\text{content}}$, a content-first metric using Hungarian assignment over blended semantic and numeric agreement. The same section establishes the metric’s discriminating power empirically and states what it does not measure.

The layers are separable by design. Layer 2b never reads a ground truth, so it cannot be tuned to one; Layer 3 never reads a document, so it cannot exploit knowledge of how a rule was extracted. Because every agent role is served by a hosted endpoint whose long generations are not reproducible under a fixed seed (Section 6.3.3), each configuration is run five times and reported as a mean with its standard deviation, rather than as a single figure that could not be distinguished from endpoint noise.

5. UNSTRUCTURED DATA PARSING STRATEGY

5.1 Overview of the Document Parser

Source documents arrive as PDFs, each encoding operational rules in a structurally distinct format: narrative prose, threshold tables, mixed maintenance text, role-zone access matrices. Before any LLM-based rule extraction can occur, these files must be converted into clean, semantically coherent plain text. Standard left-to-right PDF parsers fail this requirement: they flatten table structures into whitespace-separated tokens and discard hierarchical headings, making the output semantically incoherent for downstream extraction agents.

The Document Parser is the first agent in the pipeline. It accepts a raw SOP PDF as input and produces a sanitised plain-text file ready for the LLM extractors. It implements three sequential stages: vision-aware Markdown conversion, deterministic artifact sanitisation, and context-aware chunking.

5.2 Vision-Aware Markdown Extraction

To preserve spatial and semantic layout, the Document Parser uses IBM’s `docling` library, which applies a vision-language model to identify bounding boxes around text blocks, tables, and headers, exporting the reconstructed layout directly into Markdown. This matters because grid structure *is* content in these documents: a threshold table’s column position distinguishes a critical bound from a warning bound, and a permission matrix’s row and column labels are the only thing identifying which role may enter which zone. A text-only parser flattens both into an undifferentiated token stream, and no downstream stage can recover what was lost. For plain-text inputs (`.txt`), the parser bypasses vision extraction and reads the raw text directly, then applies the same normalisation and chunking pipeline.

5.3 Normalisation and Hallucination Scrubbing

Vision-based extraction preserves layout but introduces OCR artifacts and vision-model hallucinations. The parser executes a strict normalisation protocol (`_normalise`) to sanitise the raw Markdown before it enters the LLM context window.

5.3.1 Ligature and Encoding Correction

The source files contain directional and trend symbols that are corrupted by PDF font encoding: standalone `f i` tokens are deterministically remapped to a right-arrow (`→`) and standalone `f l` tokens to a down-arrow (`↓`), matching the original SOP semantics. The remapping is restricted to word-boundary matches, so ordinary words containing these letter pairs are left untouched. HTML entities are decoded, escaped formatting characters (e.g.

`_`) are sanitised to prevent Markdown injection errors in downstream prompts, and runs of repeated spaces are collapsed to keep the token count low.

5.3.2 Table Cell Deduplication

A recurring hallucination in layout-aware extraction occurs when overlapping bounding boxes cause cell text to be repeated sequentially (e.g., “Inspect and lubricate HIGH Inspect and lubricate HIGH”). The parser resolves this with a regex-based algorithm (`_deduplicate_table_cells`) that scans Markdown table rows (identified by pipe `|` delimiters), detects any substring that repeats itself consecutively (separated by whitespace), and collapses it to a single instance, without invoking a secondary LLM call.

5.4 Context-Aware Chunking Strategy

Splitting a document arbitrarily mid-sentence destroys semantic meaning, so segment boundaries must respect the document’s own structure. The parser splits strictly at double newlines, reliable paragraph and

section boundaries in Markdown, then merges adjacent paragraphs into a rolling buffer capped at 2 000 characters. If the final chunk is shorter than 200 characters it is merged backward into the preceding chunk to avoid uninformative microscopic segments. This guarantees that every segment is a well-formed, self-contained block of paragraphs rather than a fragment cut mid-table or mid-sentence.

The sanitised segments are then reassembled in order and saved as a single `.txt` file per document, named after its source (e.g. `SOP_001_OperatingProcedures.txt`).

Note on the two chunking stages. The chunking described above is internal to this layer: its segments are reassembled before the file is written, and their boundaries do not survive into extraction. The extraction pipeline performs its own segmentation on the reassembled text (Section 8.1), with a different size target and, critically, with every content line numbered so that each extracted record can cite the line it came from. The two are independent by design: this layer’s job is to produce faithful text, and the boundary at which a model reads that text is a decision the extraction pipeline owns, because it is the stage that must make provenance checkable.

These files serve as the direct input to both the grid search (Section 7) and the multi-agent extraction pipeline (Section 8), each of which reads the reassembled text subject to a fixed per-call character budget.

6. METHODOLOGY

This chapter details the full research methodology underlying the thesis. It describes the four corpora against which every system is evaluated, the experimental conditions under which extraction is run, the content-first metric introduced in this work together with the evidence establishing what it measures and what it does not, and the protocol that makes every reported figure a mean over repeats rather than a single draw. Individual system components (the document parser, the schema-specified grid, and the multi-agent pipeline) are described in the sections that follow; this chapter focuses on the *how* and *why* of measurement and experimental design rather than the systems themselves.

6.1 Research Design and Experimental Overview

The three layers are those introduced in Section 4 and are not restated here: document ingestion and parsing (Section 5), the schema-specified prompting grid and the schema-agnostic multi-agent pipeline (Sections 7 and 8), and evaluation by $F1_{\text{content}}$ against each corpus’s own ground truth (Section 6.6). What this chapter adds to that description is the experimental design wrapped around it: which corpora each layer is run on, under what conditions, how many times, and by what measure the results are judged.

All experiments rely exclusively on prompt engineering and architectural design decisions. No training, fine-tuning, or weight adaptation of any model was performed. This design reflects two constraints: the size of the annotated ground truths (26–86 rules per corpus) makes task-specific fine-tuning statistically untenable, and the research objective concerns what an off-the-shelf model achieves when given no corpus-specific configuration, a question that fine-tuning on one corpus would foreclose rather than answer.

6.2 Corpora: One Development Corpus and Three External Test Corpora

Extraction is evaluated on four corpora drawn from unrelated industrial domains. One serves as the development corpus, on which the pipeline was built and debugged; the remaining three are external test corpora, seen for the first time at evaluation. Table 2 summarises them.

Table 2: The four evaluation corpora. Only the development corpus was available while the pipeline was being written; the three external corpora were scored without any per-corpus configuration, prompt change, or field mapping.

Corpus	Role	Docs	GT rows	Dominant structure
Production line (iMAKS)	development	4	86	narrative, table, mixed, matrix
Biogas digestion	external test	1	47	parameter table + triggered responses
RO desalination	external test	1	26	printed rule codes + intervals
Sulfuric acid production	external test	3	70	numbered prose clauses + attribute tables

The split is the central design decision of the evaluation. A system reported only on the corpus it was developed against cannot be distinguished from one tuned to those documents, however careful the tuning was avoided in intent. The three external corpora were therefore written to differ from the development corpus in the dimensions that matter to a schema-agnostic extractor: they use different field vocabularies (`bound_low/bound_high` rather than `critLo/critHi`), different rule taxonomies, different identifier conventions (two of them number their rules with annotator-assigned codes that appear nowhere in the source text, one reuses codes printed in the document) and different granularity conventions, discussed in Section 6.5.8.

6.2.1 Development Corpus

The development corpus is the iMAKS (industrial Multi-Agent Knowledge extraction Synthetic) benchmark [5], simulating five two-shift operational days at a nine-station packaging facility. Its synthetic construction ensures that ground-truth annotations are complete and unambiguous; a real industrial SOP corpus would introduce annotation noise and confidentiality constraints orthogonal to the extraction problem studied here. Its four documents span four distinct document structures within a single corpus, which is what makes it a useful development target.

6.2.2 Facility Structure

The simulated facility consists of nine stations organised into six zones: four production stations (`ST01_FILLING`, `ST02_SEALING`, `ST03_LABELLING`, `ST04_PACKAGING`) and five auxiliary stations (`SRV01_SERVERROOM`, `WRH01_WAREHOUSE`, `CHM01_CHEMICALSTORAGE`, `RND01_RDLAB`, `CAF01_CAFETERIA`), with twenty-two physical sensors deployed across them. The station and sensor identifiers matter here only because they are the subjects that extracted rules refer to; this thesis evaluates rule extraction from the documents, not the sensor stream.

6.2.3 SOP Document Corpus and Structural Heterogeneity

The four SOP documents (SOP-001 through SOP-004) are the primary extraction target. Each encodes operational knowledge in a structurally distinct format:

- **SOP-001 (narrative):** Plain procedural prose describing operational conditions, process control actions, and zone-level safety constraints. Rules are embedded in free-form sentences with no table structure or column delimiters. The extraction challenge is purely semantic: the model must identify which sentence fragment constitutes a conditional operational constraint and parse its subject (the sensor or station), predicate (the threshold or condition), and action.
- **SOP-002 (tabular):** A sensor-threshold table with one row per sensor and four numeric threshold columns (CRIT_LO, WARN_LO, WARN_HI, CRIT_HI) plus a measurement unit column. The extraction task is more mechanical than SOP-001, amounting to column-to-field mapping, but OCR artefacts from PDF vision-extraction, unit-suffix handling, and partial rows (sensors with critical but no warning limits) introduce non-trivial failure modes. All 27 `ThresholdRule` ground-truth entries come from SOP-002, and no entry of any other class does.
- **SOP-003 (mixed):** A structured table whose rows encode maintenance procedures and correlated-fault patterns as (trigger condition, required action, priority) triples. The document is classified as *mixed* because it combines the structured tabular layout of SOP-002 with the semantic complexity of SOP-001: each row requires the model to read a multi-clause trigger condition and a multi-step action description. SOP-003 also states cross-station causal rules (RULE-CORR-01 links a sealing motor overload at ST02_SEALING_CUR to a packaging speed anomaly at ST04_PACKAGING_SPD) whose subject is a relationship between two stations rather than a single sensor. Such rules are extracted here as records like any other; whether a downstream system could act on them is outside the scope of this evaluation.
- **SOP-004 (matrix):** A role-zone access-control and occupancy matrix. Rows represent facility zones and columns represent access-permission classes or acknowledgment deadlines. Rules are encoded in individual cells rather than in sentences or table rows, requiring the extractor to correctly associate each cell value with its row label (the zone) and its column label (the permission type or deadline class). Cross-cell inference, where a model borrows a value from an adjacent cell, is the primary failure mode for this document type.

This intentional structural heterogeneity is one of the two experimental challenges. A prompt optimised for narrative prose under-extracts from tabular and matrix documents; a prompt optimised for structured rows produces poor results on prose. A pipeline that must work without being told which structure it faces cannot resolve this by routing to a structure-specific prompt, because the set of structures is not known in advance either. It is resolved instead by asking the extracting agent to copy each document’s own labels rather than to fill a fixed template, so that a table’s column headers, a matrix’s axis labels, and a sentence’s subject all become field names by the same mechanism (Section 8).

6.2.4 External Test Corpora

No external corpus took part in selecting any prompt, parameter, model, or field name; every such choice was made against the development corpus alone. The three differ, however, in how much of the pipeline’s development they were visible for, and Section 10.10 states that gradient precisely rather than describing all three as equally held out. Each comes from a domain unrelated to packaging:

- **Biogas digestion (47 ground-truth rows).** A digester operating procedure combining a parameter table (tag, equipment, measured quantity, unit, low and high bounds) with triggered-response text attached to the same physical rows. Its ground truth splits one printed table row into two annotated rows (one for the limit, one for the response) and numbers them R01, R02, codes that appear nowhere in the document.
- **Reverse-osmosis desalination (26 ground-truth rows).** A high-pressure skid procedure whose rules carry identifiers printed in the document itself (OP-ST-001), unlike the other three corpora. Its ground truth leaves the identifier column blank on rows with no printed code, which has consequences for identifier-based scoring discussed in Section 6.5.9.
- **Sulfuric acid production (70 ground-truth rows).** Three procedures for one contact-process plant — normal operation, emergency shutdown, and maintenance — covering startup sequencing, alarm and trip setpoints, shutdown response times, maintenance intervals and permit requirements. It is the only corpus of more than one document besides the development set, so it is the only external corpus that exercises the schema arbiter’s reconciliation of vocabulary across documents. Its annotation splits the two attribute tables cell by cell rather than row by row, a granularity convention whose measured cost is quantified in Section 6.5.8.

The three corpora disagree with the development corpus and with each other about what constitutes one rule. That disagreement is not incidental: it is the reason a fixed extraction schema cannot be transferred, and it bounds what any row-matching metric can report (Section 6.5.8).

6.2.5 Ground-Truth Rule Schema

The development corpus provides 86 manually annotated ground-truth rules distributed across four classes (28 operational, 27 threshold, 20 access, 11 maintenance). Its schema is described here in full because it is the corpus against which field-level placement is verified (Section 9.2); the three external corpora use their own schemas, with different field names, different class vocabularies and different identifier conventions, and none of them is supplied to the pipeline. Each development-corpus rule is a structured record with thirteen fields:

- **Identifier:** ruleId, a human-assigned naming-convention string (e.g., RULE-ST01-01, MAINT-04, RULE-THR-ST02-TMP-CRIT).
- **Categorical fields:** class, station, sensor, sensorType, severity, unit.

- **Numeric threshold fields:** `critHi`, `warnHi`, `warnLo`, `critLo`.
- **Text fields:** `condition`, `action`.

The four rule classes and their semantic roles are:

- **ThresholdRule:** numeric alarm boundaries for a specific sensor at a specific station, sourced primarily from SOP-002. The key operational content is the combination of sensor identity and numeric threshold values.
- **OperationalRule:** condition-action rules governing process control (e.g., “if filling flow drops below 105 L/min, halt the filling cycle”), sourced primarily from SOP-001. The key operational content is the textual condition and action.
- **MaintenanceRule:** scheduled or reactive maintenance procedures triggered by sensor fault patterns, concentrated in SOP-003. The key operational content is the trigger condition and the prescribed maintenance procedure.
- **AccessRule:** zone occupancy limits, access permissions, and acknowledgment deadlines, sourced from SOP-004. The key operational content is the zone identifier and the access or timing constraint.

A key characteristic of the annotations is that `ruleId` values are annotator-assigned conventions that frequently do not appear verbatim in the source SOP text. An extraction system cannot reproduce the exact annotation string without prior knowledge of the annotation convention; this is the fundamental motivation for the content-first evaluation metric defined in Section 6.5. These 86 annotated rules are the ground truth against which every development-corpus figure in this thesis is scored; the external corpora are scored against their own, described in Section 6.2.4.

6.3 Experimental Conditions

6.3.1 Inference Endpoint

Every agent role is served by Ollama Cloud through its OpenAI-compatible API, configured by `OLLAMA_BASE_URL` and `OLLAMA_API_KEY`. Nothing in the pipeline depends on the endpoint being hosted rather than local: the same code runs against any OpenAI-compatible server, and the model names are the only endpoint-specific values. The choice does, however, have one methodological consequence that shapes how every result in this thesis is reported, developed in Section 6.3.3: a hosted endpoint batches requests with other traffic, and the resulting variation cannot be suppressed by a seed.

Requests use a 600 s timeout with up to three attempts per call and a 15 s base backoff between them, and the scout and audit fan-outs issue at most four concurrent requests. The output budget is 32 768 tokens per call, generous because reasoning models bill their internal deliberation against the same budget, and a reply truncated mid-object loses every record after the cut.

6.3.2 Language Models

Two models are evaluated in the grid search (Section 7): `gemma4:31b` and `nemotron-3-nano:30b`. The multi-agent pipeline assigns `gemma4:31b` to all three agent roles: scout, inducer, and auditor.

Assigning one model to every role is a deliberate choice supported by the grid rather than a simplification. Over 100 grid runs, rescored with the metric of Section 6.5, `gemma4:31b` scored higher than `nemotron-3-nano:30b` on all ten prompting paradigms, and by a margin comparable to the spread between paradigms themselves: the spread across paradigms within `gemma4:31b` is 0.135 (0.814–0.949), while the same paradigm across the two models differs by up to 0.145. Model choice therefore matters about as much as prompting strategy, and there is no measured basis for handing any role to the weaker model. Role independence between the extracting and auditing agents (an auditor that does not share the scout’s failure modes) is available through a command-line override and is not used by default, because the only alternative measured here scored lower on every paradigm and varied more from run to run; an auditor that unreliable removes more correct records than incorrect ones.

Note on reasoning suppression. Some models expose a `reasoning_effort` parameter that gates an internal deliberation pass. `gemma4:31b` performs no such pass by default, and passing any effort value *enables* one; the parameter is therefore omitted entirely for that model rather than set to a nominal value.

6.3.3 Temperature, Seed, and the Limits of Reproducibility

All conditions run at temperature $T = 0$ with seed 42. Neither setting is sufficient to make a run reproducible on a hosted endpoint, and this thesis does not claim that it is.

Greedy decoding fixes the token choice given a probability distribution, but it does not fix the distribution. Requests to a shared endpoint are batched together with unrelated traffic, the batch composition varies between calls, and floating-point summation order varies with it. Over a short reply the resulting numerical differences are absorbed; over a long structured generation they compound until two tokens exchange rank, after which the two replies diverge outright. A fixed seed does not address this, because the randomness is not in the sampler.

Two consequences follow, and both are load-bearing for the results reported later. First, every configuration is run five times and reported as a mean with its standard deviation; a single run is one draw, and two configurations whose intervals overlap are tied rather than ranked, however different their means appear. Second, exact replay is provided by a response cache keyed on `SHA-256(model, messages)` rather than by a seed: a warm cache reproduces a previous run exactly, and any change to a prompt, a model, or a document changes the key and forces a real call, so the cache cannot silently serve a stale answer. The cache is disabled automatically whenever repeats are requested, since repeated identical requests would otherwise be served from it and the measured spread would be zero, not because the pipeline is stable, but because it never ran again.

6.3.4 Chunking and the Output Budget

Documents are not passed to a model whole. Segmentation packs blank-line-delimited blocks into chunks of approximately 3 500 characters (Section 8.1), so input length is bounded by construction and no document-level truncation limit is required; a corpus of any size is processed as more chunks rather than as a truncated document.

The output budget is set to 32 768 tokens per call. The allocation is deliberately generous for two reasons. Extracting every rule from a dense chunk as a JSON array can require several thousand output tokens, and reasoning models bill their internal deliberation against the same budget, so a model may consume most of the allowance before its answer begins. A reply cut off at the budget loses every record after the cut, which is why truncation is detected from the response’s finish reason and surfaced as a warning rather than silently absorbed (Section 8.7).

6.4 Evaluation Metric Design: Motivation and Framework

6.4.1 Limitations of Standard Exact-Match Metrics

The conventional approach to evaluating information extraction systems is to compute F1 against a ground-truth annotation using exact string matching. Applied to rule extraction, the natural analogue is identifier-based F1: a predicted rule is a true positive if and only if its `ruleId` exactly matches a ground-truth entry. Identifier-based F1 suffers from a fundamental ceiling in this setting that makes it unsuitable as an evaluation criterion, and the ceiling is a property of the annotation rather than of any system.

Ground-truth identifiers are typically annotator-assigned naming conventions absent from the source text. Three of the four corpora used here number their rules with codes that appear nowhere in the documents (RULE-ST01-01, R01, R001) while the fourth reuses codes the document itself prints. An extraction system cannot reproduce an invented convention without being told what it is, regardless of how accurately it captures the operational content, so identifier agreement on those three corpora is exactly zero for any extractor whatsoever. This is why identifier agreement is reported in this thesis as a diagnostic of the annotation convention rather than as a score (Section 6.5.9).

The consequence is a systematic evaluation blind spot: a system that extracts a semantically perfect rule, with the correct sensor, station, threshold values, and action text, may receive a false-negative because it assigned a plausible but non-matching identifier (e.g., RULE-ST01-TEMP-CRIT instead of RULE-THR-ST01-TMP-CRIT). Conversely, a system that hallucinated an incorrect threshold value but guessed the correct identifier would receive a true positive. Identifier-based F1 therefore conflates extraction quality with naming-convention alignment, which are largely independent properties.

6.4.2 Design Principles for $F1_{\text{content}}$

The primary metric introduced in this thesis, $F1_{\text{content}}$, is designed around four principles:

1. **Content primacy.** The evaluation must reflect whether the extracted rule conveys the same operational constraint as the ground-truth rule. Identifier strings are deliberately excluded from the scoring

function.

2. **Independence from both sides’ field names.** The metric must not require a correspondence between the ground truth’s field names and the extractor’s, because no such correspondence exists and constructing one per corpus would reintroduce the per-domain configuration the pipeline exists to remove. This rules out any weighting scheme defined over named fields (including class-dependent weights) and is why the comparison operates on whole-row text rather than on aligned cells (Section 6.5.3). The cost of this choice is real and is stated rather than hidden: the metric cannot tell whether a value sits in the right slot, which Section 10.2 quantifies and Section 9.2 addresses by separate means.
3. **Optimal one-to-one assignment.** Evaluation must use a globally optimal bipartite matching between the predicted and ground-truth rule sets to prevent double-counting and to produce principled precision and recall values.
4. **Partial credit with a hard TP threshold.** Rules that are mostly but not perfectly correct should receive partial credit via a continuous content-agreement score. The final true-positive decision should nonetheless be based on a threshold, to avoid inflating F1 through accumulation of many low-quality near-matches.

Why not an LLM judge? The obvious alternative to the construction above is to ask a language model whether each predicted record expresses the same rule as a ground-truth row. That design would satisfy the first two principles more directly than blob comparison does, and would probably read attribute binding, which this metric demonstrably does not (Section 10.2). It is not adopted here for three reasons specific to this study rather than for any general objection. First, reproducibility: Section 6.3.3 documents that the hosted endpoint is not reproducible under a fixed seed for long generations, so a judge built on it would make the *measuring instrument* as variable as the system measured, and the run-to-run movement reported in Section 9.1 could no longer be attributed to the pipeline rather than to the judge. The metric used here is deterministic: rescoring the same predictions reproduces every figure in this thesis exactly. Second, independence: the pipeline’s scouts, inducer and auditor are served by gemma4:31b, and a judge drawn from the same family would share its blind spots, exactly the objection raised against the shared-model auditor in Section 8.5. Third, auditability: every acceptance decision here is a number a reader can recompute from the audit trail (Section 6.7), whereas a judge’s verdict can be inspected but not recomputed. The two approaches are complementary rather than exclusive, and the audit trail was written in a form that makes the comparison executable: the stratified blinded pair sample of Section 10.2 can be labelled by a judge as readily as by a person. Neither labelling has been collected, so no claim is made here about which agrees better with human judgment.

6.5 The Metric: Content Agreement over Optimally Assigned Pairs

The evaluation compares two sets of records that share no vocabulary. A ground-truth row names its fields `critLo`, `bound_low`, or `min_bound` depending on which corpus wrote it; a predicted record names its fields whatever the source document called them. No mapping between the two exists, and constructing one by

hand for each corpus would reintroduce exactly the per-domain configuration the pipeline exists to remove. The metric therefore aligns nothing by name.

6.5.1 The Procedure End to End

Before the individual components are defined, Algorithm 3 states the whole procedure, so that each subsequent subsection can be read as an expansion of one numbered step rather than as an isolated definition.

Table 3: Scoring one prediction file against one ground truth. Steps 1–4 are independent of the acceptance threshold, which is why a threshold sweep (Section 6.6) reuses a single execution of them.

Input: ground-truth rows G , predicted records P , threshold τ , weights $w_{\text{sem}}, w_{\text{num}}$	
1.	Detect and remove the identifier column and any provenance columns from G ; remove the pipeline’s fixed bookkeeping columns from P . Detection is structural (Section 6.5.4).
2.	Collapse every remaining row of G and record of P into a text blob by joining its surviving values with “ ”.
3.	Extract the numbers appearing anywhere in each blob (Section 6.5.5).
4.	For every pair $(g, p) \in G \times P$, compute $\text{agree}(g, p)$ by Equation 1, forming an $ G \times P $ matrix.
5.	Solve the matrix as a linear assignment problem, maximising the total agreement over all one-to-one pairings. This yields $\min(G , P)$ pairs.
6.	Count a pair as a true positive when $\text{agree}(g, p) \geq \tau$.
7.	Set $\text{FP} = P - \text{TP}$ and $\text{FN} = G - \text{TP}$, then compute precision, recall and $F1_{\text{content}}$.
8.	Emit one audit row per decision (matched pairs, unpaired ground-truth rows, unpaired records) recording every intermediate quantity (Section 6.7).

Two properties of this procedure are worth stating explicitly because they are frequent sources of confusion. First, *no field is ever compared to a field*: the comparison operates on whole-row blobs, so a ground truth calling a quantity `bound_low` and a record calling it `ll` are never required to agree on the name. Second, *the comparison is not lexical*: two blobs sharing no words can still be paired if their sentence embeddings are close, and two blobs sharing many words can fail to reach the threshold. The word-level overlap reported in the audit trail is a diagnostic for the reader and plays no part in steps 4–7.

6.5.2 A Worked Example

The following is a real comparison, taken verbatim from the audit trail for run 4 of the biogas corpus, and is reproducible by opening that file.

A pair that is counted. Ground-truth row R04 and predicted record SOPBDG003MID-002 produce these two blobs after step 2:

```
GT AlarmControlResponse | DGT-101 | pH | <6.5 at any time |
   DCS stops feed pump PMP-201, raises Critical alarm
   DGT-101 pH Low-Low | Critical | 6.5 | pH
PRED OperatingLimit | If pH <6.5 at any time, DCS stops feed pump
```

```
PMP-201 and raises Critical alarm "DGT-101 pH Low-Low". |
Primary Digester 1 | 7.5 | 6.8 | pH | DGT-101 | -
```

Note what the two sides disagree about. The ground truth calls this rule an `AlarmControlResponse`; the record calls it an `OperatingLimit`, a label from a vocabulary the inducer derived for this corpus and which the ground truth never uses. The record also carries the equipment name and the normal operating band (6.8–7.5) that the ground-truth row omits, and the two sides order their fields differently, because each follows its own column order. Under exact matching on any field this pair fails. The metric proceeds as follows.

Step 3, numbers. The ground truth yields $N_g = \{-101, 6.5, -201, -101, 6.5\}$ and the record $N_p = \{6.5, -201, -101, 1, 7.5, 6.8, -101\}$. The negative values are asset tags: the number extractor reads the hyphen in DGT-101 as a minus sign, so the tag contributes -101 on both sides (Section 6.5.5 discusses this).

Step 4, agreement. Each ground-truth number finds an exact counterpart in the record, so $\text{cover}(N_g, N_p) = 1.000$. The reverse direction is weaker: the record’s 1 (from “Primary Digester 1”) answers to nothing in the ground-truth row, and its 6.8 and 7.5 are the unannotated operating band, so $\text{cover}(N_p, N_g) = 0.854$. Their harmonic mean is $\text{numOverlap} = 0.921$. The sentence embeddings give a cosine of 0.836. Therefore

$$\text{agree} = 0.6 \times 0.836 + 0.4 \times 0.921 = 0.870,$$

which is the value the audit trail records, along with the per-number pairing (6.5→6.5 (1.000), and so on) that produced each direction.

Steps 5–6. The assignment selects this pair, and $0.870 \geq 0.6$, so it is a true positive. For orientation the audit also notes that the two blobs share words at a Jaccard overlap of 0.667, a number that played no part in the decision.

A pair that is rejected. The assignment is exhaustive over $\min(|G|, |P|)$ pairs, so it pairs rows even when no good partner remains. Ground-truth row R22 was assigned to record SOPBDG003MID-017:

```
GT   NormalOperatingLimit | FLR-601 | Flame Temperature |
      shall be at or above minimum | 850.0 | C
PRED AccessRestriction | FLR-601 | No personnel entry is permitted
      without explicit authorisation from the Shift Supervisor
```

Both rows concern the same asset, FLR-601, which is why the assignment paired them once better partners were taken. But they state entirely different kinds of rule, and the cosine is only 0.231. The numeric term is instructive here: every number the record carries is justified, so $\text{cover}(N_p, N_g) = 1.000$, but the record carries only the asset tag and misses the 850 the ground truth states, so $\text{cover}(N_g, N_p) = 0.500$ and their harmonic mean is 0.667. The blended score is $0.6 \times 0.231 + 0.4 \times 0.667 = 0.405$, well below $\tau = 0.6$. The pair is recorded as `REJECTED` (below threshold) and counts as a false negative for R22 and a false positive for the record. This is the correct outcome, and it illustrates why both directions of the numeric term are needed: a record cannot clear the threshold merely by carrying nothing unjustified.

From decisions to the reported figure. That run produced 28 records against 47 ground-truth rows. The assignment formed 28 pairs, of which 26 cleared the threshold and 2 did not, leaving 19 ground-truth rows unpaired for want of a record. Step 7 then gives

$$TP = 26, \quad FP = 28 - 26 = 2, \quad FN = 47 - 26 = 21,$$

$$P = \frac{26}{26+2} = 0.929, \quad R = \frac{26}{26+21} = 0.553, \quad F1_{\text{content}} = \frac{2 \times 0.929 \times 0.553}{0.929 + 0.553} = 0.693.$$

Every quantity in this derivation is written to `f1_derivation_<corpus>.csv` with the substitution shown, so the arithmetic can be checked without re-running anything.

6.5.3 Why There Is No Field-by-Field Comparison

The ground truth has fields, and the natural expectation is that scoring compares them one by one: the ground truth's `bound_low` against the record's corresponding value, its `action` against the record's `action`, and so on. No such comparison is performed, and this subsection states precisely why and what is done instead, because it is the single most consequential design decision in the evaluation.

The obstacle. A field-by-field comparison requires a correspondence between the two sides' field names. That correspondence does not exist. The biogas ground truth names its bounds `bound_low` and `bound_high`; the desalination ground truth names them `min_bound` and `max_bound`; the development corpus names them `critLo`, `warnLo`, `warnHi` and `critHi`. On the predicted side the names are whatever each source document printed: `l1` and `h1` in the example below. Building the correspondence means a human reading both schemas and writing a mapping per corpus, which is exactly the per-domain configuration the pipeline exists to remove; a metric requiring it could not be applied to a new facility any more than a schema-specified prompt could.

What is done instead. Both sides are flattened. Every surviving column's *value* is concatenated in column order; the *names are discarded entirely*. The comparison then happens between two flat strings. For the ground-truth row R04 of the worked example (Section 6.5.2), the twelve stored columns reduce as follows:

<code>id</code>	<code>= R04</code>	<code>-> dropped (identifier)</code>
<code>category</code>	<code>= AlarmControlResponse</code>	<code>-> kept</code>
<code>asset</code>	<code>= DGT-101</code>	<code>-> kept</code>
<code>tag</code>	<code>=</code>	<code>-> dropped (empty)</code>
<code>parameter</code>	<code>= pH</code>	<code>-> kept</code>
<code>condition</code>	<code>= <6.5 at any time</code>	<code>-> kept</code>
<code>action</code>	<code>= DCS stops feed pump ...</code>	<code>-> kept</code>
<code>severity</code>	<code>= Critical</code>	<code>-> kept</code>
<code>bound_low</code>	<code>= 6.5</code>	<code>-> kept</code>
<code>bound_high</code>	<code>=</code>	<code>-> dropped (empty)</code>
<code>unit</code>	<code>= pH</code>	<code>-> kept</code>
<code>source</code>	<code>= SOP-BDG-003</code>	<code>-> dropped (provenance)</code>

leaving the blob AlarmControlResponse | DGT-101 | pH | <6.5 at any time | DCS stops feed pump PMP-201, ... | Critical | 6.5 | pH.

The predicted record is flattened by the identical rule. It has its own columns, discovered from the source document rather than declared anywhere, and for this corpus there are 28 of them, one for every field any record in the corpus was observed to carry. The record matched to R04 populates eight:

```

id           = SOPBDG003MID-002  -> dropped (bookkeeping)
category     = OperatingLimit      -> kept
action_remarks = If pH <6.5 at any time, DCS ... -> kept
equipment    = Primary Digester 1  -> kept
hl           = 7.5                 -> kept
ll           = 6.8                 -> kept
parameter    = pH                  -> kept
tag          = DGT-101             -> kept
unit         = -                   -> kept
(19 further discovered columns, empty for this record -> dropped)
source_file   = SOP_BDG_003_...txt -> dropped (bookkeeping)
source_span   = ...:L21#1 | ...    -> dropped (bookkeeping)

```

Note that the two sides express the same bound under different names (the ground truth's bound_low against the record's ll, taken from the document's own column header) and that neither name survives into the blob.

The consequence, stated plainly. After flattening, the metric no longer knows that 6.5 was a bound_low. It knows only that 6.5 appears in the ground-truth row and that some number close to it appears in the record. Two things follow, and both are real:

- **Semantic flexibility.** A record is not penalised for naming a field differently from the annotator, for splitting one field into two, or for assigning a category from a vocabulary the ground truth never uses. In the worked example the two sides disagree on the category outright (AlarmControlResponse against OperatingLimit, a label the inducer derived for this corpus) and the pair is still correctly counted, because the operational content agrees.
- **Loss of positional binding.** A value placed in the *wrong* field produces the same token in the blob as one placed correctly. If a record attached 6.5 as an upper bound rather than a lower one, the flattened strings would be indistinguishable. The metric therefore checks that the right values are present, not that each is bound to the right slot. This is not a conjecture: permuting quantities between records (so that every rule carries another rule's bound) costs -0.001 F1 (Table 5), which is to say the metric does not detect it at all. The mechanism is the assignment itself, behaving as designed: the Hungarian step is free to re-pair each record with whichever ground-truth row its post-permutation content now resembles, so the damage is absorbed by re-assignment rather than priced. Freezing the assignment before perturbing would measure a stricter property than the metric possesses; the figure reported is the one that applies to a metric which assigns optimally. The blind spot is carried into the limitations as Section 10.2.

The design is therefore a deliberate trade: field-name independence, which is what makes cross-corpus evaluation possible at all, is bought at the price of field-level precision. A metric that recovered the latter without reimposing a per-corpus schema mapping would be a genuine improvement, and none is offered here.

This framing has a limit, and the results should not be read past it. Throughout, surplus records are attributed to granularity rather than to error, and the evidence supports that: the development corpus’s unpaired records state facts the documents contain, and regrouping the sulfuric acid annotation moves its figure by 0.147 without changing a prediction. But “not an error” is not the same as “not a problem”. A pipeline that emits 131 records where an annotator wrote 86 imposes a real cost on whatever consumes them: a knowledge graph populated from that output carries redundant nodes, a human reviewing it reads the same rule stated three ways, and deduplication becomes a downstream task that someone must specify. The granularity argument establishes that the surplus is not fabrication. It does not establish that the surplus is free.

6.5.4 Blob Construction (Steps 1–2)

The operation is easiest to state as an example. A ground-truth row `{min_temp: 50, unit: C}` and a predicted record `{low_limit: 50, uom: C}` describe the same rule in two vocabularies that share no column name. Collapsed, both become `50 | C`, and they compare as identical. That is the whole idea: the column names are discarded and only what they held is compared, which is what allows a corpus that invented its own field names to be scored against an annotation that invented different ones.

Concretely, each ground-truth row and each predicted record is collapsed into a single text blob: every column’s value, whatever the column is called, joined in order with “ | ”. Two kinds of column are removed first, and both are identified structurally rather than from a list of known names, so that a corpus using an unfamiliar column name still resolves correctly:

- **The identifier column.** Every column is scored by $u - \bar{\ell}/200 + h$, where u is the fraction of non-empty values that are distinct, $\bar{\ell}$ is the mean value length, and h is a bonus of 0.05 if the column name is one of `id`, `ruleId`, `rule_id`. Columns with $u < 0.9$ are disqualified outright, and the highest-scoring survivor is taken. The criterion encodes the statistical signature of an identifier (distinct on almost every row, and short) while the name bonus is small enough to break a near-tie and never to decide one alone. The column is excluded from the blob because matching identifiers measures naming convention rather than content (Section 6.5.9).
- **Provenance columns.** A column is provenance if it holds the same value on every row, or if it has at most $\max(6, \lfloor 0.1 |G| \rfloor)$ distinct values *and* each of those values is at most 24 characters, contains no space, and contains a digit: the shape of a document code rather than of rule content. The second test exists because a corpus of four documents gives its source column four values, which the constant-value test misses; a genuine categorical field such as `severity` fails it because its values are words. On all four corpora exactly one column is removed by this test (`source`) and no content column is affected.

Both thresholds above are declared parameters rather than inline constants, and are listed with the other free parameters in Table 4.

Each finished blob then passes through one fixed normalisation step, applied identically to both sides. Documents and annotations routinely encode the same character differently: a typesetter emits the non-breaking hyphen U+2011 inside an asset tag where an annotator types an ASCII hyphen, or the true minus sign U+2212 where an annotator types the same ASCII hyphen. Nothing semantic distinguishes the spellings, but every reader downstream of the blob treats them as distinct tokens, and the numeric extractor of Section 6.5.5 reads a minus sign in one and not in the other. Dash variants are therefore folded to the ASCII hyphen, non-breaking and typographic spaces to the ASCII space, and zero-width marks removed. The fold is a character-level identity, not a scoring decision: it is fixed before any corpus is scored, is applied to ground truth and prediction alike, and cannot move a score in a direction that depends on which side used which encoding. It is deliberately narrower than the standard NFKC normalisation, which additionally rewrites subscripts and superscripts as digits and would manufacture numbers that neither side wrote (CO₂ becoming C02), corrupting the very term the metric relies on.

On the four corpora reported here the fold is inert: rescoreing every corpus with normalisation disabled reproduces all four figures exactly (0.719, 0.715, 0.850, 0.649). It is retained not because it earns anything on this evidence but because the condition it guards against is a property of how a document was typeset rather than of the extractor, and a corpus exhibiting it would otherwise be penalised for an encoding difference its annotator never saw. Reporting a normalisation step that changes nothing is the honest outcome of testing it; the step is declared in Table 4 so that a reader can verify the claim by turning it off.

The predicted side drops the pipeline’s own fixed bookkeeping columns by the same principle: `id`, `source_file`, and `source_span` (plus `chunk_id`, accepted defensively although the saved output does not carry it), the columns that hold a record’s identity and provenance rather than extracted content. In particular the recorded source span, which names the line a record was read from, is excluded: no ground truth carries such a column, so every token it contributes is unmatched by construction and can only dilute the comparison.

6.5.5 Agreement Between Two Blobs

For a ground-truth row g and a predicted record p with blobs b_g, b_p :

$$\text{agree}(g, p) = w_{\text{sem}} \cdot \cos(\text{SBERT}(b_g), \text{SBERT}(b_p)) + w_{\text{num}} \cdot \text{numOverlap}(g, p), \quad (1)$$

with $w_{\text{sem}} = 0.6$ and $w_{\text{num}} = 0.4$. The semantic term is cosine similarity of sentence embeddings [20] (all-MiniLM-L6-v2), which is what allows `EnvironmentalLimit | TMP | 175 | 195` to align with `ThresholdRule | TMP` although neither side knows the other’s vocabulary.

The numeric term. The numeric term exists because sentence embeddings barely distinguish 175 from 195, and a rule with the wrong bound is wrong. Numbers are extracted from a blob by the pattern `-?\d+\.\d*` applied to the whole string, so a value is found wherever it was written and whichever column held it. Writing N_g and N_p for the numbers so obtained:

$$\text{cover}(A, B) = \frac{1}{|A|} \sum_{x \in A} \max_{y \in B} \max \left(0, 1 - \frac{|x - y|}{\max(|x|, |y|)} \right). \quad (2)$$

The score for one number is therefore 1 for an exact match and decays with relative error, reaching 0 only when the nearest value on the other side is of opposite sign or more than twice as far from zero. The per-number pairing that produced the average is written to the audit trail, so a reader can see which predicted value was credited against which ground-truth value and how close it had to be.

The two sides are then combined symmetrically, as an F1 over numbers:

$$\text{numOverlap}(g, p) = \frac{2 \cdot \text{cover}(N_g, N_p) \cdot \text{cover}(N_p, N_g)}{\text{cover}(N_g, N_p) + \text{cover}(N_p, N_g)}. \quad (3)$$

Both directions are required, and the reason is not symmetry for its own sake. $\text{cover}(N_g, N_p)$ alone asks only whether the record contains each of the ground truth’s values; it prices surplus values at zero, so a record can improve its score by carrying numbers it never extracted. That is not a theoretical concern but a measured one: appending to every predicted record the numbers already present in *other records of the same file* — changing no extracted content, and using no knowledge of the ground truth — raises the recall-only figure on all four corpora, by +0.021 on biogas to +0.178 on sulfuric acid. Under Equation 3 the same padding costs between 0.030 and 0.177 instead. A metric that pays for padding is not measuring numeric correctness, whatever it charges for corrupting a value, so the symmetric form is the one reported throughout.

The second direction has a cost that must be stated alongside it. Its precision half charges a record for carrying values the annotation omits, and some of those values are correct extractions of facts the ground truth simply does not record (Section 10.5). Tolerance scoring keeps the charge mild, since a surplus value near an annotated one still scores highly, and the measured effect on the reported figures is small — the largest movement across the four corpora is +0.023 — but the charge is real, and it is the price of pricing padding at all.

One consequence of this extraction is visible in the worked example of Section 6.5.2 and is declared as a parameter rather than resolved silently: a hyphen inside an asset tag reads as a minus sign, so DGT-101 contributes −101. It contributes −101 on both sides, because the normalisation of Section 6.5.4 has already reconciled the two encodings of that hyphen; what remains open is not which side reads a minus but whether such values should count at all. Such values are about 15% of all numbers extracted from these ground truths. They are counted, because a shared asset tag is genuine evidence that two rows concern the same equipment; the choice is exposed as a parameter (Table 4) so that it is auditable rather than buried in a regular expression. Rescoring every corpus with these values excluded measures what the choice costs: the development corpus does not move at all, biogas rises slightly from 0.715 to 0.725, sulfuric acid falls from 0.649 to 0.601, and desalination rises from 0.850 to 0.917. The two larger moves have different sources. Sulfuric acid’s instrument tags are printed in the documents and reused verbatim by the annotator, so there the tag-derived values are shared evidence and discarding them discards signal. Desalination’s ground truth repeats its printed rule codes inside the annotated rule text where the pipeline’s content fields do not, so there the tag-derived values are one-sided and cost the reported figure 0.067. Counting them therefore

lowers the reported figure on the corpus where the choice most favours the system, which is why it is the setting reported throughout: the alternative would raise the headline number on desalination by 0.067 on the strength of a scoring decision rather than an extraction one.

The conceptual objection to counting them at all should be conceded rather than argued away. A hyphen read as a minus sign is an artifact of the number extractor, not a designed feature: an asset tag is an identifier, and DGT-101 is not the quantity -101 in any sense the document intends. That a shared tag is genuine evidence two rows concern the same equipment is true, but it argues for entity agreement being its own term in Equation 1, scored separately and on its own scale, rather than for smuggling identifiers into a term that exists to check thresholds. The present treatment is defensible because it is symmetric, declared, and measured in both settings, and because the setting reported is the one that costs the system rather than favours it; it is not defensible as a design. Separating entity matching from numeric agreement is left to future work, and until it is done the numeric term should be read as mixing two kinds of evidence that happen to share a notation.

When a ground-truth row contains no numbers at all the numeric term is undefined rather than zero, and the agreement is the semantic cosine alone. This asymmetry is reported in the audit trail rather than hidden, because it means a row without numbers must reach the acceptance threshold on semantic similarity unaided, while a row with numbers may clear it on a weaker semantic match carried by numeric agreement.

6.5.6 Optimal Assignment and $F1_{\text{content}}$

Pairing is global, not greedy. The full $|G| \times |P|$ matrix of agreement scores is solved as a linear assignment problem by the Hungarian algorithm [18], maximising total agreement over all one-to-one pairings. Each ground-truth row is therefore credited to at most one record and each record to at most one row, so a single well-formed record cannot be counted as satisfying three different rules.

A pair counts as a true positive when its agreement reaches the threshold $\tau = 0.6$. Writing n_G and n_P for the number of ground-truth rows and predicted records:

$$TP = |\{(g, p) \text{ assigned} : \text{agree}(g, p) \geq \tau\}|, \quad FP = n_P - TP, \quad FN = n_G - TP, \quad (4)$$

with precision, recall and $F1_{\text{content}}$ following in the usual way. Note that the assignment maximises *total* agreement rather than the *count* of pairs above threshold; the two can differ in principle, and the conventional choice is taken here.

6.5.7 Free Parameters

Every constant the evaluator uses is declared in one configuration object rather than written inline, because each is a researcher degree of freedom: a number a human chose, which could in principle have been chosen by looking at results on the corpus being scored. Collecting them makes them countable, and every one is overridable from the command line so that a reported figure can be shown stable across a range rather than dependent on one setting.

Table 4: Every free parameter of the evaluation, its value, and what it governs. The defaults are conventional starting points, not values fitted to any ground truth in this study; Section 6.6 reports where the operating point sits on the threshold curve.

Parameter	Value	Governs
τ (threshold)	0.6	agreement required for an assigned pair to count as a true positive
w_{sem}	0.6	weight on the SBERT cosine term
w_{num}	0.4	weight on the numeric-agreement term
numeric term	symmetric	whether numeric agreement is an F1 over both sides’ numbers or the ground truth’s recall alone (Equation 3)
embedding model	all-MiniLM-L6-v2	sentence embeddings; English-only, matching the corpora
id uniqueness minimum	0.9	distinct-value fraction below which a column cannot be the identifier
id length penalty divisor	200	how strongly long values disqualify a column from being the identifier
id name-hint bonus	0.05	tie-break nudge for a conventional identifier name
provenance distinct floor	6	maximum distinct values for a document-label column
provenance distinct frac.	0.1	row-fraction alternative to the floor; the larger of the two limits applies
provenance max value len.	24	maximum length of a value that still looks like a document code
drop tag-derived numbers	off	whether to discard negative values arising from hyphenated asset tags

6.5.8 The Precision Ceiling Imposed by Granularity

Equation 4 has a consequence that must be stated before any result is read. The assignment forms at most $\min(n_G, n_P)$ pairs, so whenever a system emits more records than the ground truth has rows, the surplus are false positives *by construction*: they had no row left to be assigned to, whether or not their content is correct. Precision is therefore capped at

$$\text{precision} \leq \frac{\min(n_G, n_P)}{n_P}. \quad (5)$$

This is not a defect of the implementation but a property of one-to-one row matching, and it interacts directly with the corpora’s disagreement about what constitutes one rule (Section 6.2.4). A system whose granularity differs from the annotator’s is penalised for the difference, in either direction.

The mismatch does *not* reduce to the pipeline being uniformly more verbose. On the development corpus it runs both ways at once. For threshold rules the ground truth is the *finer* of the two: its 22 threshold sensors are annotated as 50 rows (ten sensors carry three rows each, eight carry two) because one printed table row is split into an operational band, a critical trip, and a threshold record. The pipeline emits one row per sensor there, and is therefore coarser. Elsewhere (matrix cells, inter-station dependencies) the pipeline

is finer. The two effects net out to more records than rows overall, but no single collapse or split rule could reconcile them, which is why none is applied. The ceiling is reported alongside the precision it caps in every results table, since a precision of 0.595 against a ceiling of 0.656 and the same figure against a ceiling of 1.0 describe entirely different systems.

The sulfuric acid corpus allows the size of this effect to be measured directly rather than argued, because there the disagreement is isolated to two tables and can be undone without touching the predictions. Both tables have *attribute* columns — headers naming different kinds of information about one subject (a detection setpoint, an automatic action, a required operator response, a deadline), which are not interchangeable and could not be swapped — and the annotation records one row per *cell* rather than per table row. The three threshold rows of the operating procedure’s parameter table become twelve annotated rules, and the four trip scenarios of the emergency procedure become sixteen, four of which consist of nothing but a duration (1 minute, 2 minutes) with no subject attached. The pipeline reads those tables the other way: for the sulfur feed tag it emits a single record carrying the normal band, the low alarm, the high alarm and the trip action as four separate fields, so every printed value is captured and the record is nonetheless scored as one true positive against four annotated rows.

Regrouping only those two tables to one row per table row — 70 annotated rows becoming 49, with no change whatever to the predictions being scored — raises $F1_{\text{content}}$ from 0.649 ± 0.049 to 0.796 ± 0.065 , and recall from 0.552 to 0.796. The difference of 0.147 is the price of a single annotation convention, paid by a system that extracted the same content either way. Two things follow. First, the 0.649 reported in Table 7 is the figure against the annotation as delivered, and it is the figure this thesis reports: regrouping a ground truth after seeing how a system reads it is exactly the adjustment that makes a benchmark unfalsifiable, and the regrouped variant is retained in the repository as a labelled diagnostic rather than as a result. Second, and more generally, a 0.147 swing from a convention that no metric of this family can observe places a floor on how finely any cross-corpus comparison of these numbers should be read — larger than the gap between two of the four corpora in Table 7.

6.5.9 Identifier Reuse as a Diagnostic, Not a Score

Identifier agreement is reported, but deliberately not as a precision/recall/F1 triple. What it measures is whether the corpus’s annotator reused codes printed in the source document or invented their own numbering, which is a property of the annotation and not of the system under test. A ground truth numbering its rules R01 and R001 (strings appearing nowhere in the document) yields zero identifier agreement for any extractor whatsoever, while one reusing printed codes such as OP-ST-001 yields high agreement for any extractor that copies what it reads.

Expressed as an F1 and placed beside $F1_{\text{content}}$, this quantity invites the reading that the system’s true score is near zero on the corpora where it is zero. It is therefore reported as a fraction of the ground truth’s distinct identifiers recovered verbatim, labelled as a diagnostic, with the distinct-identifier count given alongside, necessary because a ground truth leaving its identifier column blank on some rows has fewer identifiers than rows, and the fraction would otherwise be read against the wrong denominator.

6.6 Establishing That the Metric Measures Anything

A metric built on sentence embeddings can report a respectable score while discriminating nothing, because embeddings place any two documents of the same genre at a non-trivial cosine similarity. A reported $F1_{\text{content}}$ of 0.75–0.99 is therefore not self-interpreting, and this section establishes what it is worth by measurement rather than by argument. All figures come from the same evaluator that produces the headline results, run over the same five repeats per corpus, with deterministic seeded perturbations.

6.6.1 The Null Case

The decisive test is whether the metric collapses when the predictions cannot possibly be right. Each corpus’s predictions are scored against every *other* corpus’s ground truth: nothing is correct, so a metric reading content must fall towards zero, while one reading genre similarity will not.

Pooled per corpus, the null score is 0.014–0.037, far below every reported figure. The worst individual pairing is higher and must be reported as such: development-corpus predictions scored against the biogas ground truth reach 0.101, and sulfuric acid predictions against the same ground truth 0.084. The practical consequence is that reported figures should be read against a floor of roughly 0.10 rather than against 0; the floor measured on each ground truth individually, and the effect of subtracting it, are given in Section 10.3.

The symmetric numeric term of Equation 3 is what makes this floor as low as it is. Under the recall-only alternative the same test returned 0.000–0.193, with a worst pairing of 0.193: a foreign corpus’s records were not charged for the numbers they carried that the ground truth never mentions, and a large annotation therefore offered the assignment enough accidental agreement to reach a fifth of a usable score. Pricing unjustified values roughly halves the floor, which is the clearest evidence that the second direction is doing work rather than adding symmetry for its own sake.

6.6.2 Perturbation Tests

Each perturbation damages one aspect of the predictions, so the score it costs is attributable to that aspect. Table 5 reports the mean $F1$ lost across the four corpora.

Table 5: Mean $F1_{\text{content}}$ lost under targeted degradation of the predictions. A perturbation that costs nothing marks something the metric does not check, and is reported as a limitation rather than omitted. A negative entry means the score rose slightly under the perturbation; movements within ± 0.01 are inside run-to-run noise and read as “not detected”.

Perturbation	F1 lost	Interpretation
Every numeric value corrupted	0.201	bound values are genuinely verified
Records padded with the corpus’s other numbers	0.097	surplus values are charged, not free
Bounds reversed <i>within</i> each record	0.000	slot binding is <i>not</i> checked at all
Numeric values permuted between records	−0.001	value-to-rule binding is <i>not</i> checked
Category label permuted between records	+0.007	the discovered label carries almost no weight
Token order scrambled (vs. control)	0.000	word order is <i>not</i> read

The first two rows are the ones that license the metric’s use for rule extraction, and they are complementary. Replacing every number with a different number costs 0.201 on average, from 0.085 on the corpus where it costs least to 0.324 on the corpus where it costs most: bound values are not decoration in this score. Padding each record with the numbers carried by other records of the same file — a perturbation that adds no wrong values, only unjustified ones, and uses no knowledge of the ground truth — costs a further 0.097 on average. The two together say that the metric charges both for stating a value wrongly and for stating one it cannot support, which is what distinguishes verifying numbers from merely finding them.

The third row is the sharpest statement of what this metric cannot do, and it deserves to be read before the others. Reversing the order of a record’s own numeric values — so that a row printing a critical low bound of 0.65 and a critical high bound of 1.05 now reports 1.05 as its low bound and 0.65 as its high — costs **exactly zero** on every one of the four corpora, to four decimal places. This is not the assignment recovering by re-pairing, as it is in the row below: the multiset of values inside each record is untouched, so there is nothing for the assignment to recover from. The agreement function simply cannot see which slot a value occupies, because the blob it compares retains the values and discards the names. A plant configured from records perturbed this way would alarm at the wrong end of every range, and $F1_{\text{content}}$ would report the corpus unchanged.

The consequence must be stated without softening: *no figure in this thesis is evidence that an extracted value sits in the correct slot*. The headline metric measures whether the right content was recovered, which is a necessary condition for a usable extraction and not a sufficient one. Slot correctness is a separate property, it is measured separately in Sections 9.2 and 9.2.1, and it is measurably imperfect where it can be measured at all.

The remaining rows are limitations of the same family, and together they are more restrictive than the first two rows alone suggest. Taking another record’s quantities while keeping one’s own subject and prose (the failure in which the right rule carries the wrong bound) costs nothing at all (−0.001; the score in fact rises slightly, by 0.017 on the development corpus): the assignment simply re-pairs, recovering by reallocation what the corruption removed. Permuting the discovered category label between records costs almost nothing (+0.007), because after flattening the label is one column among several dozen. And scrambling token order, measured against an order-preserving control that isolates the cost of the rewrite itself, costs 0.000.

The summary is therefore narrow: $F1_{\text{content}}$ verifies that the right *values* are present somewhere in the extracted set, and verifies essentially nothing else: not which record holds them, not the label attached to them, not the order in which they are written. It is a set-membership measure over values, and should be read as one. This also explains why the numeric term of Equation 1 is load-bearing: it supplies the only discrimination the metric reliably has.

6.6.3 Where the Operating Point Sits

The acceptance threshold $\tau = 0.6$ is a human choice and therefore a researcher degree of freedom. What matters is not that it is defensible in the abstract but where it sits on the curve. Sweeping τ from 0.4 to 0.9 shows that it is a local maximum on *no* corpus: there is no corpus where 0.6 scores better than both 0.55 and 0.65, which is the shape a threshold chosen to flatter the system would have. All four curves decline through

the operating point, so on every corpus a lower threshold would report a higher figure — at $\tau = 0.4$ the four report 0.793, 0.747, 0.917 and 0.810 against the 0.719, 0.715, 0.850 and 0.649 reported at 0.6. That the reported operating point is the more conservative choice on all four corpora is the relevant property here.

The largest movement any corpus shows across a single 0.05 step near the operating point is 0.023, on desalination; the development corpus moves 0.015, biogas 0.016 and sulfuric acid 0.020. The ranking of corpora by $F1_{\text{content}}$ is nonetheless not stable across the full sweep, so no claim that one corpus is intrinsically harder than another is supported by these results.

The semantic weight is swept by the same discipline. Holding $\tau = 0.6$ and varying w_{sem} from 0.4 to 0.8, the development corpus reports *higher* with more numeric weight (0.747 at $w_{\text{sem}} = 0.4$, against 0.719 at the default) while biogas, desalination and sulfuric acid all report higher with more semantic weight (0.725, 0.887 and 0.722 at $w_{\text{sem}} = 0.8$, against 0.715, 0.850 and 0.649). The corpora pull in opposite directions, so no single weighting flatters all four, and the default of 0.6/0.4 is a peak on none of them.

6.7 Reviewability: The Per-Record Audit Trail

A summary figure cannot be checked. Alongside every score the evaluator emits one row per comparison decision: every assigned pair, then every ground-truth row and every record left unpaired. Table 6 lists what each row records. A separate file writes out the counts and the substituted precision, recall and F1 arithmetic per run, as shown at the end of Section 6.5.2.

Table 6: Columns of the per-record audit trail. Everything above the rule is used by the metric; everything below it is reported for the reader and plays no part in any score.

Column	Content
decision	TP, REJECTED (below threshold), UNPAIRED GT, or UNPAIRED RECORD
counts_as	how the decision enters Equation 4
gt_row, gt_id	the ground-truth row and its identifier
pred_row, pred_id	the paired record and its identifier
semantic_cosine	the SBERT term of Equation 1
numeric_overlap	the numeric term, or blank when the row has no numbers
numeric_recall_gt_found	$\text{cover}(N_g, N_p)$: the ground truth’s values found in the record
numeric_precision_pred_justified	$\text{cover}(N_p, N_g)$: the record’s values justified by the row
combined_score	the blended agreement
score_formula	the arithmetic written out, e.g. $0.6 \cdot 0.855 + 0.4 \cdot 1.000 = 0.913$
threshold	the τ in force
numeric_detail_gt_to_pred	which predicted number was credited against which ground-truth number, with its score
numeric_detail_pred_to_gt	the same pairing in the reverse direction, behind the precision half
gt_numbers, pred_numbers	the extracted number sets
runner_up_score, margin_over_runner_up	the best score obtainable from any <i>other</i> record, and the margin
gt_text_compared, pred_text_compared	the two blobs, verbatim
word_overlap_jaccard	lexical overlap of the two blobs (<i>diagnostic</i>)
shared_words	tokens present in both (<i>diagnostic</i>)
gt_only_words, pred_only_words	tokens unique to each side (<i>diagnostic</i>)

Two properties make this an audit rather than a second implementation. First, the trail is emitted from the same assignment object the metric is computed from, so it cannot drift from the reported figure; recomputing F1 from the audit files alone reproduces every reported value exactly. Second, each ground-truth row’s *runner-up* is recorded (the best score obtainable from any other record) so a reader can see whether a pairing was decisive or a near-tie. Across the 945 assigned pairs of all twenty runs, 82.5% of rows receive their own best-scoring record and 17.5% yield it to another row that the global optimum needs it for; 16.8% of pairs sit within ± 0.05 of their runner-up and should be read as close calls rather than confident matches.

For orientation the audit also reports a word-level view of the same two blobs (shared tokens and Jaccard overlap) explicitly marked as diagnostic and excluded from the score. It is included because the first question asked of any semantic metric is whether it is really comparing text word by word, and the answer is only convincing when the word-level picture is visible next to the score it is being distinguished from.

7. SCHEMA-SPECIFIED EXTRACTION: THE PROMPTING GRID

This section establishes what single-prompt extraction achieves when the target schema *is* known, and, equally important, what that achievement is contingent upon. The grid serves two purposes in this thesis.

It selects the prompting paradigm carried into the multi-agent pipeline’s scout agents (Section 8.2), and it provides the contrast of Section 9.3 against which the cost of requiring no schema is measured.

7.1 What the Prompt Is Given

The distinguishing property of every condition in this grid is that its prompt carries the development corpus’s answer format. Specifically, the prompt supplies:

- the exact output schema, naming all thirteen ground-truth fields (`ruleId`, `class`, `station`, `sensor`, `sensorType`, `condition`, `action`, `severity`, `critHi`, `warnHi`, `warnLo`, `critLo`, `unit`);
- the rule taxonomy, naming the ground truth’s classes explicitly and instructing the model which fields to populate for each;
- the identifier convention, by example (`RULE-ST01-01`, `RULE-THR-ENC07-LUX-CRIT`);
- worked examples of complete output records, using real station and sensor identifiers from the benchmark family;
- an explicit column-to-field mapping for the threshold tables (`CRIT_LO` $\rightarrow$ `critLo`, and so on).

This is a strong and entirely legitimate configuration for a system deployed on a corpus whose schema somebody has already written down. It is reported here as such. What it cannot do is transfer: applying any of these conditions to the biogas, desalination, or sulfuric acid corpora requires a human to read those documents and rewrite all five elements above. The grid therefore measures schema-specified extraction, not general extraction, and Section 9.3 treats it accordingly.

7.2 Models and Experimental Parameters

The grid evaluates two models, `gemma4:31b` and `nemotron-3-nano:30b`, across ten prompting paradigms with five repeats per condition, for 100 runs in total. Both are served through the same OpenAI-compatible endpoint as the multi-agent pipeline (Section 6.3.1), so the two systems differ in what their prompts contain and in their architecture, not in the model or the serving stack.

Network resilience is handled by an exponential backoff mechanism allowing up to three attempts per call, backing off from a 15-second base delay. For all paradigms except one the temperature is 0 with a fixed seed of 42; the self-consistency paradigm uses a temperature of 0.3 across three independent inner runs by construction, since it requires diverse reasoning paths. As established in Section 6.3.3, neither setting makes a run bit-reproducible on a hosted endpoint, which is why every condition is reported as a mean over five repeats with its standard deviation.

7.3 Prompting Paradigms and Complexity Levels

To systematically assess how LLMs interpret industrial logic, the grid search categorises prompting strategies into four distinct complexity levels:

- **Level 0 (Direct Extraction):** Includes the naive approach, which enforces a strict JSON schema without additional constraints. The `few_shot_static` paradigm enhances this by providing two concrete examples one `OperationalRule` and one `ThresholdRule` illustrating the required output format and numeric field mapping. Finally, the `graph_informed` strategy introduces mandatory constraints, forcing the LLM to map its outputs to specific sensor naming conventions (e.g., `ST01_FILLING_TMP`).
- **Level 1 (Reasoning Before Action):** Introduces intermediate cognitive steps before JSON generation. The `cot_basic` paradigm requires standard step-by-step reasoning, while `cot_structured` enforces a strict five-step analytical process: (1) enumerate all stations and zones, (2) identify the sensors and measurement types per station, (3) extract threshold conditions with their numeric values, (4) identify causal dependencies between stations, and (5) identify access-control, occupancy, and maintenance rules. The `pre_act` paradigm requires the model to summarise the document’s structure, the categories of rules present, and any cross-section dependencies before producing the extraction.
- **Level 2 (Iterative Refinement and Ensemble Voting):** Incorporates multi-turn logic and ensemble strategies. The `self_consistency` paradigm runs three independent extractions at temperature 0.3 and then aggregates the results through a score-based voting mechanism (Section 7.4). Both `reflexion` and `reflexion_guided` use the `graph_informed` system prompt as their conversational base, inheriting its mandatory sensor-naming constraints, whereas `self_consistency` uses the simpler naive base without those constraints. The `reflexion` strategy utilises a two-turn loop in which the model critiques its own initial output for missing rules, imprecise or incorrect numeric conditions, wrong sensor and station names, and duplicate or poorly-structured entries. The `reflexion_guided` paradigm extends this to three turns, adding a rigid domain-specific checklist in the final turn that covers six categories: (1) completeness across all stations, (2) cross-station dependencies, (3) temporal anomaly patterns (`SPIKE`, `DRIFT`, `STUCK`, `OUT_OF_RANGE`, `CORRELATED`), (4) predictive and corrective maintenance protocols (`MAINT-01` to `MAINT-08`), (5) access-control and zone-occupancy rules, and (6) correctness of all numeric threshold values.
- **Level 3 (Tool Use / ReAct):** The `react_abox` paradigm allows the model to actively verify sensor names against a localised Knowledge Graph (ABox). The LLM can halt its generation to emit a `VERIFY: <sensor_name>` command. The system evaluates this request using fuzzy token-overlap matching and returns either a `SENSOR_OK` confirmation or a `SENSOR_NOT_FOUND` warning with up to three syntactically similar suggestions. The loop is bounded to a maximum of five turns; if this limit is reached without a final JSON response, the last assistant message is parsed directly.

7.4 Ensemble Voting and Content Agreement

The `self_consistency` paradigm (Level 2) relies on an ensemble voting mechanism to aggregate the results of three independent LLM calls. To determine whether two independently generated rules describe

the same operational constraint, a `_content_agreement` algorithm calculates a weighted similarity score (from 0.0 to 1.0).

Categorical fields (`class`, `station`, `sensor`, `sensorType`, `severity`, `unit`) are evaluated using exact string matching after normalisation, while text fields (`conditions` and `actions`) are evaluated using a Jaccard token-overlap approximation. Numeric fields within `ThresholdRule` instances are subjected to a tolerance-based similarity calculation to prevent minor rounding differences from invalidating a match. Field weights are non-uniform: among categorical fields, location identifiers (`station`, `sensor`) carry weight 2.0, rule class (`class`) carries weight 1.5, and the remaining categorical fields (`sensorType`, `severity`, `unit`) each carry weight 1.0. Primary numeric bounds (`critHi`, `critLo`) carry weight 3.0, secondary bounds (`warnHi`, `warnLo`) carry weight 1.5, and text fields carry weight 0.5 for `ThresholdRule` entries (where numerics are the primary signal) or 2.0 for all other rule classes.

The system implements a `score_based` voting strategy by default. Rather than requiring rules to share an identical `ruleId`, this strategy builds a pairwise agreement matrix across all rules generated in the ensemble. If the similarity score meets or exceeds the predefined threshold of 0.55 (i.e. ≥ 0.55), the rules are clustered via greedy merging in descending agreement order. A rule is only retained for the final output if its cluster spans at least 2 out of 3 independent runs. For retained `ThresholdRule` clusters, a secondary majority vote is applied across the threshold-related fields (`critHi`, `critLo`, `warnHi`, `warnLo`) and the measurement unit (`unit`) to resolve minor threshold discrepancies and produce a single stabilised value.

7.5 System Resilience and Output Parsing

Given the tendency of LLMs to deviate from strict schemas, especially under high token loads, the system includes extensive fallback mechanisms. Experiment state is preserved via a JSON registry that marks runs as `partial` upon initiation and `completed` only upon successful disk write (with atomic write-then-rename saves), allowing interrupted grid searches to resume seamlessly; a per-run partial CSV checkpoint is additionally written after each document. A dedicated heartbeat thread logs a timestamp and the current experiment identifier every 30 seconds, so silent hangs during prolonged API calls are distinguishable from normal long-running inference, and a signal handler persists the registry on interruption so completed runs are never lost.

To extract valid data from malformed LLM responses, a cascading four-strategy JSON parser is deployed. Internal `<think>...</think>` reasoning traces are stripped before any parsing attempt. If standard parsing fails, the system attempts to repair truncated JSON strings by balancing unclosed brackets. If the output contains Markdown artifacts, regex extraction is used to isolate the data blocks. As a final fallback, the parser iterates through the corrupted output to salvage any individually well-formed rule objects.

8. MULTI-AGENT EXTRACTION PIPELINE ARCHITECTURE

The pipeline’s design follows from a single constraint: it is given a directory of operating documents and nothing else. It does not receive a target schema, a rule taxonomy, a list of field names, a worked example, or any indication of whether a document is prose, a table, or a matrix. Every design decision below is a

consequence of that constraint, and each is stated with the failure it avoids.

The graph is

$$\text{load_segment} \rightarrow [\text{scout}]^* \rightarrow \text{induce} \rightarrow \text{assemble} \rightarrow [\text{audit}]^* \rightarrow \text{save},$$

where starred stages fan out over the document in parallel and the remainder run once. Figure 1 shows the graph as it is actually constructed, and the rest of this section expands one stage per subsection.

step2_multi_agent_generic.py — LangGraph pipeline
no target schema, no rule taxonomy, no worked example is supplied to any stage

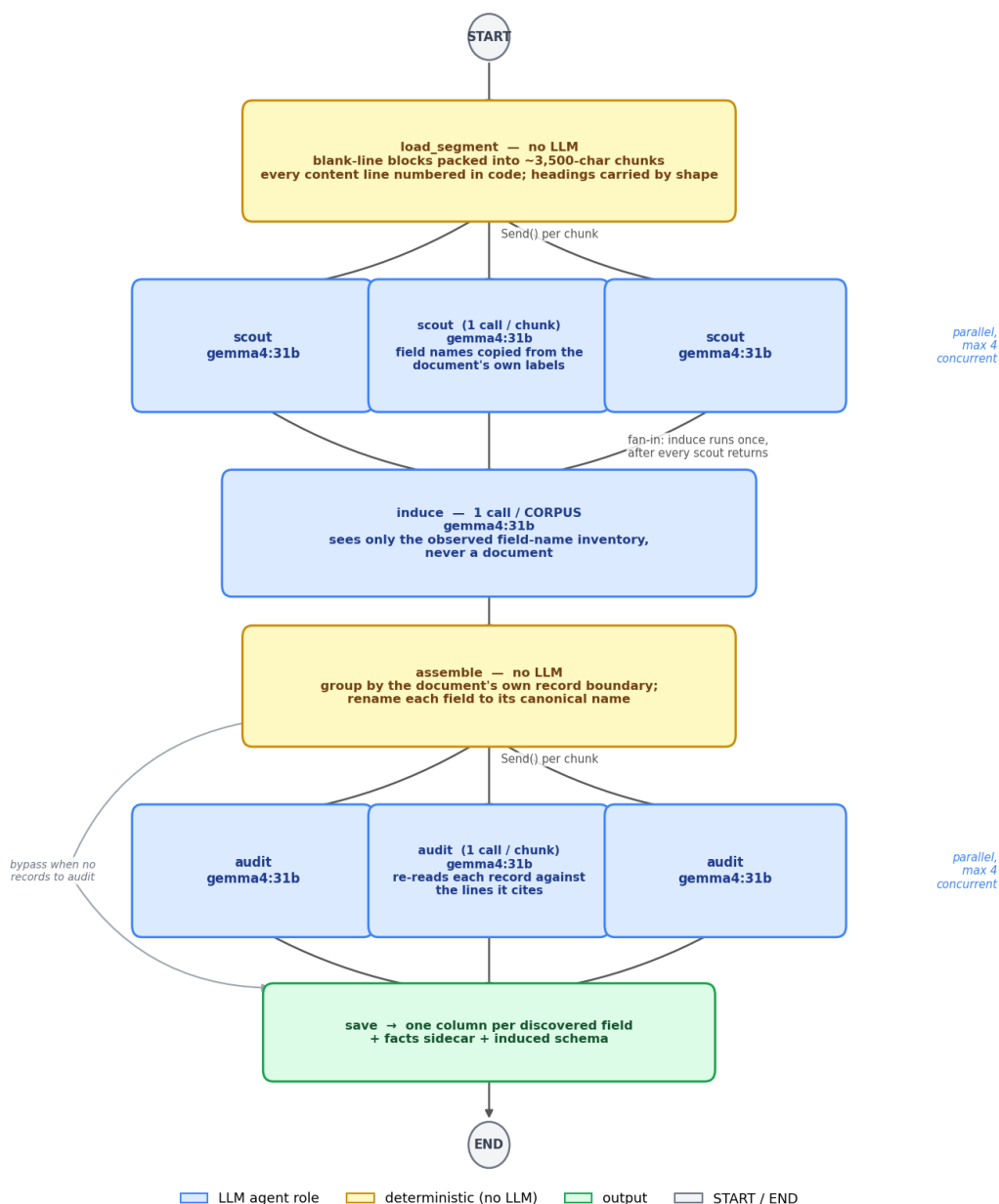


Figure 1: The LangGraph pipeline. Blue nodes are LLM agent roles, yellow nodes are deterministic, green is output. Starred stages in the text correspond to the three-wide rows here: `scout` and `audit` are instantiated once per document chunk and dispatched concurrently, so the three boxes denote “one of these per chunk” rather than exactly three. The grey edge on the left is the conditional bypass taken when no chunk produced records to audit. The figure is generated from the pipeline module’s own constants, so the model names, the chunk size and the concurrency limit shown cannot drift from the code that runs.

Three properties of the topology are worth reading off the figure before the stages are described individually, because each is a design decision rather than an implementation detail.

Two of the six stages use no language model at all. Segmentation and assembly are deterministic code (yellow). This matters for what can be claimed about the output: the boundaries at which text is cut, the line numbers records cite, the grouping of facts into rows, and the renaming of fields to their canonical form are all reproducible operations that do not vary between runs. The non-determinism discussed in Section 6.3.3 enters only at the three blue stages.

The two fan-out stages are per chunk; the inducer is per corpus. Scout and audit scale with document length, so their cost is linear in the corpus and they run at a bounded concurrency. The inducer runs exactly once no matter how large the corpus is, because its input is not the corpus but the inventory of field names the scouts reported. The narrow waist in the middle of the figure is therefore not a drawing convention: it is the one point where the pipeline takes a corpus-wide view.

The scout → induce edge is a fan-in, not a pass-through. Every scout branch must return before the inducer runs, since an arbiter shown one chunk’s vocabulary could not reconcile it with another’s. This is the only synchronisation barrier in the graph, and it is why the inducer cannot be merged into the scout stage however tempting the saving in calls.

8.1 Stage 1: Segmentation and Line Numbering

Segmentation uses no LLM. Each document is split on blank lines into blocks, blocks are packed into chunks of approximately 3 500 characters, and, most consequentially, every content line is assigned a number by code.

Numbering lines in the harness rather than asking the model to track provenance is what makes grounding checkable later. An agent is only ever asked to cite a line number it is currently looking at, which is among the most reliable things a language model can be asked to produce; the alternative, asking a model to reproduce the text it drew from, invites paraphrase and cannot be verified automatically.

Section headings are detected by shape alone (a short, isolated, unpunctuated line) and carried into every chunk they cover. Without this, the answer to “which asset does this table belong to” is lost at the first chunk boundary, because the heading naming the asset sits in the preceding chunk. Detection by shape rather than by a pattern of known heading formats is again a consequence of the no-schema constraint.

8.2 Stage 2: Scout Agents

One LLM call per chunk, issued in parallel with a concurrency of four. The scout reads a numbered chunk and emits one record per rule *the document itself delimits*: a printed rule identifier where the document prints one, otherwise one table row, one matrix cell, or one rule-stating sentence.

Two properties of the scout prompt carry the schema-agnosticism:

- **Field names are copied from the document’s own labels.** A table’s column headers become the record’s field names; a matrix’s axis labels become its subject and field name; a sentence contributes the quantity it names. The scout is never given a set of fields to fill, so it cannot force a document into

a foreign shape, and a corpus whose rules genuinely carry fourteen distinct slots yields fourteen fields because fourteen were observed.

- **No worked example is supplied.** An example demonstrates one particular document shape, and a model shown one reproduces that shape on documents that do not have it. This is the mechanism by which a schema-specified prompt achieves high accuracy on the corpus it was written for and transfers poorly; omitting the example gives up the former to avoid the latter.

The prompt ends with a chain-of-thought instruction: reason through the chunk, then emit JSON. This is the `cot_basic` paradigm, and its selection from the grid of Section 7 is constrained before it is empirical. Only three grid paradigms outscore it, and two of them cannot be carried into a schema-agnostic scout at all: `reflexion_guided`’s final turn is a corpus-specific checklist, and `reflexion` inherits mandatory sensor-naming constraints from its `graph_informed` base, so in both cases the margin is bought with exactly the corpus knowledge this pipeline is defined not to have. The third, `pre_act`, does transfer and is statistically tied with `cot_basic` (0.894 ± 0.009 against 0.892 ± 0.005); `cot_basic` is preferred for its lower run-to-run variance, and because `pre_act`’s preliminary survey of document structure has little to survey on the single chunk a scout reads.

8.3 Stage 3: Schema Induction

One LLM call per corpus, and the only stage with a global view. The inducer never sees a document. Its input is the inventory of field names and category labels the scouts actually produced, with occurrence counts and sample values, and its task is to merge synonyms into one corpus-level schema.

Deriving the schema from what the corpus was observed to contain, rather than proposing one in advance from a document’s opening pages, is what allows the field count to follow the evidence. It also resolves the problem that two documents in one corpus may name the same role differently: reconciliation happens here, once, so that the pipeline’s output is expressed in a single vocabulary even though its scouts spoke several.

Whether that reconciliation improves agreement with a given ground truth is a separate question from whether it is desirable, and the two can diverge: an annotator’s field names are the scoring target, so renaming a field to a canonical form that differs from the annotator’s reduces measured agreement even where the canonical form is the better output. This stage is therefore included on output-quality grounds, and no claim is made here about its effect on $F1_{\text{content}}$.

What can be measured without a field-name ground truth is whether the induction behaves as a stable function of the corpus or as a fresh invention on every run, and the five production repeats answer this, because each run saved its induced schema. Stability tracks how much discretion the documents leave the scouts. The development corpus is the most stable of the four: 34–44 canonical fields per run, a mean pairwise Jaccard overlap of 0.813 between the runs’ field sets, a core of 31 of 50 union fields present in every run, and the same choice of all three role pointers on every repeat. The three external corpora leave the scouts more discretion and the induced vocabulary varies correspondingly (mean pairwise Jaccard 0.447–0.476), around a smaller core (9 of 47 union fields on desalination, 15 of 71 on biogas, 19 of 86 on sulfuric acid), and on those corpora the role pointers are not always chosen identically across repeats. The variation originates

upstream rather than in the arbiter: the inducer sees only the inventory that run’s scouts reported, so what varies is what was observed, not what was reconciled. This inherited vocabulary variation is also precisely what a field-name-sensitive score would convert into noise; the content-first metric’s indifference to field names is what keeps the reported spread at ± 0.043 or below despite a canonical vocabulary that overlaps run-to-run by less than one half on three corpora.

Withholding the documents from this stage is deliberate. An arbiter that could read the source text would be able to introduce fields no scout reported, which would make the schema a product of the arbiter’s priors rather than of the corpus. Restricting it to the observed inventory means the schema cannot contain anything the corpus did not exhibit.

What this stage’s cost scales with. Because the inducer reads the inventory rather than the corpus, its input is bounded by the number of *distinct* field names and category labels the scouts reported, not by the number of documents or their length. On the development corpus, four documents totalling about 13,500 characters yield 449 extracted facts but only 36–51 distinct field names and 11 distinct category labels across the five runs, which with counts and sample values is a prompt of a few hundred tokens. Every other stage is per-chunk and parallel, so this single global call is the only place where a corpus-wide quantity enters one context window.

That bound is a property of the architecture; whether it stays small on a facility-scale corpus is not established by four documents, and the growth observed here does not yet flatten. Accumulating the distinct field names document by document on the development corpus gives 9, 22, 28, 41 in one representative run: still rising at the fourth document. The inventory therefore grows with vocabulary diversity, and the point at which vocabulary diversity saturates for a real facility is unmeasured. What can be said is that the quantity at risk is the inventory rather than the text, that it is one to two orders of magnitude smaller than the corpus at this scale, and that its format (a bag of names with counts) admits hierarchical merging of per-document inventories should it ever exceed a context window. No such merging is implemented or tested here, and the scale limitation is restated in Section 10.8.

One implementation detail must be declared alongside that argument, because it would bind before the context window does. The inventory handed to the arbiter is truncated to the 120 most frequent field names. On the corpora evaluated here the cap never engages — the largest inventory any single run produced is 44 names — so no reported figure is affected by it. But the truncation is by frequency, which means the first names discarded at scale would be the rarest ones, and those are precisely the names the arbiter’s own instructions tell it to preserve: a field used by three records out of two hundred is the only place those three records’ information lives. Past roughly 120 distinct field names the implementation would therefore contradict its own specification, silently and without error, discarding rare vocabulary before the arbiter is ever consulted. This is a property of the current implementation rather than of the design — the cap exists to bound one prompt’s size and could be replaced by the hierarchical merge described above — but it is the concrete point at which this stage stops scaling, and it is nearer than the context window.

No claim of scalability is made anywhere in this thesis, and this is the reason. The pipeline is demonstrated on nine documents across four corpora; what happens at a facility holding thousands of procedures

is untested, and the paragraph above identifies the specific mechanism that would fail first and the specific class of information it would discard — the rarest fields, which in a safety setting are frequently the ones carrying edge-case triggers. Naming that mechanism is not the same as having solved it.

The shape a scalable version would take. The route out is described here because a limitation is more useful when the reader can see what would remove it, and it is described as a sketch because none of it is implemented or evaluated in this thesis. The single global call would become a reduction over a tree. Each document’s scouts emit a local inventory; inventories are merged pairwise, each merge reconciling exactly two vocabularies and emitting canonical names together with the aliases that were folded into them; the merged result is itself an inventory, so the operation composes, and the root sees a bounded number of canonical names rather than the union of every raw one. The number of rounds is logarithmic in the document count, and the input to any single call is bounded by the width of one merge rather than by the size of the corpus, which is the property the current implementation lacks.

Two things would have to hold for that to be an improvement rather than a rearrangement. Retaining the alias list at every level is what keeps the mapping from a document’s own label to the canonical name recoverable, without which the provenance guarantee of Section 8.1 would not survive the merge. And rarity would have to stop being the discard criterion: a name should be removed only when a merge finds it synonymous with a name that is kept, never because it was infrequent, which is the precise inversion of what the 120-name truncation does today. Whether pairwise merging preserves vocabulary as well as one global view does is an empirical question this thesis does not answer.

8.4 Stage 4: Deterministic Assembly

No LLM. Records are grouped by the document’s own record boundary, each field is renamed to its canonical name from the induced schema, and one row per rule is emitted. The boundary is defined concretely. Where the document prints an identifier for a rule (a rule code such as OP-ST-001, a coded table row), that identifier is the boundary: two scouted readings carrying the same printed identifier merge into one record, which is what reunites a rule restated on both sides of a chunk boundary. Where no identifier is printed, the boundary is the physical source line together with the order the record was read in, and id-less records are never merged, because one line may legitimately state several distinct rules (a matrix line holds one per cell) and a redundant record is recoverable downstream whereas a silently merged one is not. Granularity is thus decided by evidence printed in the document rather than by any convention the pipeline carries.

This is the stage that gives rise to the granularity mismatch analysed in Section 6.5.8. Extraction proceeds at the finest granularity the document supports (one atomic fact per cell or value) because that choice loses no information, whereas a coarser choice cannot be undone. Whether a given ground truth annotates at that same granularity is outside the pipeline’s knowledge, and the metric charges the difference as error.

8.5 Stage 5: Grounding Audit

One LLM call per chunk, again in parallel. The auditor re-reads each assembled record against the numbered lines that record cites, and drops or corrects anything not actually stated there. This stage targets numeric

hallucination: a threshold the model invented is not present on the line it claims to come from, and the citation created in Stage 1 is what makes that discrepancy checkable in principle.

One safeguard applies. A whole-chunk rejection (an auditor claiming that every record from a chunk is ungrounded) is treated as an audit failure rather than as a finding whenever the chunk carried three or more records, and the records are retained. The reasoning is a declared design assumption rather than a measured frequency: a verdict rejecting every record in a sizeable chunk is more plausibly a failure of the audit call itself — a malformed or truncated reply, or a verdict list that did not follow the requested form — than a correct judgement that a whole section of an operating document states nothing. Without the guard, one bad audit response silently deletes a document’s entire contribution. For a chunk carrying only one or two records the guard does not apply and the rejection stands, because at that size rejecting everything is indistinguishable from the auditor doing its job on a record or two, and no inference of malfunction is available.

The foregoing states the stage’s purpose and mechanism. It is not a claim that the stage raises $F1_{\text{content}}$: what the audit targets is whether a record asserts something its cited line does not support, and Section 10.2 establishes that the metric used here checks that property only weakly. How often the stage intervenes is therefore measured directly instead. The pipeline writes every auditor verdict to a per-run sidecar file, so the rate is read off the reported batch itself rather than inferred: across the 20 runs behind Table 7, 1176 records were audited in 80 chunk audits, *none* was dropped, and 12 were corrected — an intervention rate of 1.02%. No chunk-level rejection override was triggered on any run. The rate is not uniform across corpora: it is 0.15% on the development corpus and 0% on desalination, against 5.71% on biogas and 1.22% on sulfuric acid, so what the stage does depends on the document rather than being a constant property of the configuration. The aggregation is produced by a script over the retained sidecars and is in the accompanying repository.

Three conclusions follow. The first bounds the stage’s effect on the reported figures, the second says what the stage nonetheless does, and the third restrains what may be inferred from either.

First, the arithmetic isolates which stages drive the reported performance, and it does so without an ablation run. A record’s contribution to $F1_{\text{content}}$ can be altered by the audit only if the audit drops that record or changes any part of its content. Zero were dropped and 12 of 1176 were corrected, so at most 1.02% of records differ from what assembly produced. Inspecting those twelve is informative in itself: eight rewrote the field `lines`, which holds a record’s own provenance rather than any rule content, and only four altered a field carrying extracted content. The eight are not excluded from the bound — the defect described below puts that provenance into the scored blob, so they could in principle move a score — but rescoring with the field removed shows they do not, which leaves the auditor’s substantive intervention across the entire reported batch at four records in 1176.

What these figures establish is attribution. The extraction performance reported in Table 7 is produced almost entirely by the scout and assembly stages, with the audit contributing negligibly to $F1_{\text{content}}$ on these corpora; the headline numbers therefore stand on the scouts’ and assembly’s output essentially unedited. The same bound disposes of the converse worry — that the precision in Table 7 might be an artifact of aggressive silent filtering — since an auditor that removes nothing filters nothing.

Second, the stage is not inert, and the per-corpus rates are what show it. An intervention rate of 0% on desalination against 5.71% on biogas is not noise around a constant; it is the stage responding to different documents differently, which is the behaviour a conditional safeguard should exhibit. What it never did, on any corpus, was drop a record.

The eight provenance rewrites noted above are a defect rather than a behaviour, and are declared here because the audit log in the accompanying repository shows them plainly. The pipeline maintains a set of internal keys — a record’s identifier, its category, its cited lines, its source file — which are bookkeeping rather than extracted content, and the scout stage filters them out of the fields it reports. The auditor’s correction path does not apply the same filter, so a verdict returning a correction keyed `lines` writes that record’s own line numbers into its content, where they become a column of the output and enter the scored blob. Because the metric extracts numbers from that blob, the concern is specific: injected line numbers are numbers, and the numeric term carries 40% of the agreement score. The effect was therefore measured rather than presumed. It reaches eight records of one corpus across the twenty reported runs — three each on two biogas runs and two on a third — and rescoring those runs with the field excluded returns 0.725 ± 0.012 against the reported 0.715 ± 0.023 . The defect therefore costs the biogas figure 0.011, carried entirely by the two runs where the injected field is populated, each of which loses 0.027.

That cost is itself a consequence of the numeric term being symmetric (Equation 3), and the comparison is worth stating because it is the clearest demonstration of what the second direction buys. Under the recall-only alternative the same defect costs *exactly nothing*: the injected line numbers are surplus values, a recall-only term prices surplus values at zero, and the corpus scores 0.725 ± 0.012 whether the field is present or not. A metric that cannot see a record asserting numbers its source line never contained is not one that should be trusted to detect a hallucinated threshold either, and this defect — introduced by accident — is the closest thing to a controlled injection test these corpora provide.

The direction of the bias should be stated plainly: the reported biogas figure is *depressed* by 0.011 relative to what the pipeline would score without the defect, so it is conservative rather than flattering. The repair is one line — the correction path must apply the same internal-key filter the scout path already applies — and it is not applied here because doing so changes extraction output, which requires re-running every corpus, and a fresh batch differs from this one by the endpoint’s own run-to-run variation: up to 0.049 on the corpora reported here. Re-running would move every figure in this thesis by more than the defect it removes, and in an unknown direction, so the defect is disclosed and bounded instead, and carried in the code with a comment recording all of the above. It should be fixed together with the next full re-run, when the cost of re-measurement is being paid for other reasons anyway.

Third, and restraining both of the above, a drop count of exactly zero across 1176 audited records is consistent with two different explanations, and the measurement alone does not separate them: either there was almost nothing to find, or an auditor served by the same model as the scout cannot see what the scout got wrong. The two are not equally supported here. The field-level verification of Section 9.2 is an independent check on the same records, performed against the source documents by deterministic code rather than by a model, and on the three runs where a correspondence can be established it finds every bound cell carrying the printed value in the printed column; the only cells where the pipeline and the annotation disagree are

four where the annotation is wrong. On those runs there was in fact little for the auditor to catch, which is what its intervention rate of 0.15% would predict. That argument reaches one corpus and one failure mode. It does not show that a shared-model auditor would catch hallucinations produced by a scout that had, for whatever reason, misread a noisier document: a checker drawn from the same weights as the writer is a correlated check, and correlated checks fail together. Role independence is available as a command-line override (Section 8.6); measuring whether it raises the intervention rate would require a second model of comparable strength, which this study does not have, and injecting hallucinations deliberately to see whether the stage catches them is an experiment these corpora do not pose.

8.6 Model Assignment

All three agent roles are served by gemma4:31b, for the reason given in Section 6.3: over 100 grid runs it outperformed the alternative on all ten paradigms, by a margin comparable to the spread between paradigms, and there is no measured basis for assigning any role to the weaker model. Role independence between scout and auditor is available through a command-line override; it is not the default because the only alternative measured here was both weaker and less stable, and an unreliable auditor removes more correct records than incorrect ones.

8.7 Error Resilience

Three failure modes are handled explicitly, each because it was observed to occur against a hosted endpoint rather than anticipated in the abstract.

Empty content from a reasoning model. A model that returns its deliberation in a separate field and its answer in the content field can exhaust the output budget before the answer begins, yielding an empty content string with a length-based finish reason. Parsing that as an empty result would silently record “this chunk contained no rules”. It is instead raised and retried.

Truncation. A reply cut off at the output budget loses every record after the cut. This is detected from the finish reason and surfaced as a warning rather than absorbed, since the symptom otherwise appears as an unexplained drop in record count.

Transient endpoint failures. Calls are attempted up to three times with exponential backoff from a 15 s base. The timeout is deliberately generous at 600 s: reasoning models with a large output budget legitimately take minutes, and a shorter timeout discards valid work in progress.

9. RESULTS

Every figure in this section is a mean over five repeats of the same configuration, reported with its standard deviation, for the reason given in Section 6.3.3: a single run against a hosted endpoint is one draw, and two configurations whose intervals overlap are tied rather than ranked. Precision is reported alongside the ceiling that Equation 5 imposes on it, since the two are not comparable across corpora without it.

9.1 Extraction Across Four Corpora

Table 7 reports the pipeline on the development corpus and on the three external corpora, none of which contributed to any configuration choice. No prompt, parameter, field mapping, or threshold was changed between the four; the pipeline was invoked identically in each case, differing only in the input directory. The corpora do differ in how much of the pipeline’s development they were visible for, which Section 10.10 sets out.

Table 7: Multi-agent pipeline, five repeats per corpus. Rows are ordered as in Table 2: the development corpus first, then the three external corpora in the order they were introduced. The ordering is deliberately independent of the scores, so that the table cannot be read as leading with its best result. *Max P* is the ceiling that the record count imposes on precision (Equation 5); precision must be read against it rather than against 1.0.

Corpus	Role	GT	Records	$F1_{\text{content}}$	P	Max P	R
Production line (iMAKS)	dev	86	131.0 ± 0.0	0.719 ± 0.000	0.595	0.656	0.907
Biogas digestion	external	47	28.0 ± 0.0	0.715 ± 0.023	0.957	1.000	0.570
RO desalination	external	26	27.2 ± 0.4	0.850 ± 0.019	0.831	0.956	0.869
Sulfuric acid	external	70	49.0 ± 1.0	0.649 ± 0.049	0.788	1.000	0.552

Four observations follow. The first fixes the reference point, the second is the central result, and the last two constrain how the table should be read.

The development corpus, where the pipeline was built, scores 0.719 ± 0.000 . This is the figure every other number in the table should be compared against, because it is the only corpus whose documents were available while the pipeline was being written. It is reported first for that reason rather than because of where it falls in the ranking. Its recall of 0.907 is the highest of the four and its precision of 0.595 the lowest, a pattern the second and third observations explain.

Performance does not fall away on unseen material, but neither is it uniform. The three external corpora score 0.715, 0.850 and 0.649 against the development corpus’s 0.719. One sits above it, one essentially level with it, and the third, sulfuric acid, sits 0.070 below. The honest reading is therefore not that unseen corpora are handled as well as the development corpus in every case, but that the development corpus holds no privileged position: it is neither the best nor the worst of the four, which is the pattern a system shaped around its evaluation data would not produce. The spread across the four corpora, 0.649 to 0.850, is wider than any single corpus’s run-to-run variation, so the differences between corpora are real and demand explanation rather than averaging. The two observations that follow supply it, and in both cases the explanation is annotation granularity rather than extraction error.

The development corpus is precision-limited by granularity, not by error, and one observation demonstrates this outright. Across the five dev repeats both the number of true positives and the number of records emitted are *identical*: 78 of 86 ground-truth rows matched from 131 records, every run, giving $F1 = 0.719$ with zero variance. The 131 records against 86 annotated rows put a ceiling of 0.656 on precision before a single comparison is made, and the observed 0.595 must be read against that ceiling rather than

against 1.0. The same rules were found each time and only their itemisation differs from the annotation’s; $F1_{\text{content}}$ charges that difference as error. The unpaired records state facts the source documents contain and the annotation schema has no row for, such as the inter-station dependencies printed in a table the ground truth does not cover; every value they assert appears in the document they cite, which Section 10.5 measures rather than assumes.

Variance differs sharply across corpora, and a zero standard deviation does not mean determinism. Sulfuric acid’s standard deviation of 0.049 (range 0.610–0.733) is the largest of the four; biogas’s is 0.023 and desalination’s 0.019. The development corpus returned an identical 0.719 on all five repeats, and biogas an identical record count on all five. Neither is evidence of a deterministic pipeline. The five runs behind each figure are genuinely independent calls — the response cache is disabled automatically whenever repeats are requested, precisely so that a warm cache cannot replay one run five times and report the result as zero variance — and the underlying prediction files differ from one another in every case. What converges is the score, not the output: on the development corpus all five runs recovered the same 78 ground-truth rows from the same number of records, so the differences between them fell entirely in field naming and category labelling, which $F1_{\text{content}}$ is designed not to charge for. A corpus can therefore report a zero standard deviation and still be varying underneath, and a batch of five repeats should not be read as capturing the full spread of a hosted endpoint that is non-deterministic for long generations. This is why two configurations whose intervals overlap are treated as tied throughout this thesis.

9.2 Field-Level Verification Against the Source Documents

$F1_{\text{content}}$ deliberately ignores field identity (Section 6.5.3), so it cannot answer the question a downstream consumer actually needs answered: is each value in the right slot? A knowledge graph populated from these records needs the critical low bound to be the critical low bound, not merely a number that appears somewhere in the row. This section answers that question directly, outside the metric, by comparing cells against the ground truth on the one corpus where a field correspondence can be established without inventing one.

The correspondence is obtained by normalising both sides’ column names (case folding and separator removal) which maps the ground truth’s `critLo`, `warnLo`, `warnHi`, `critHi` onto the pipeline’s `crit_lo`, `warn_lo`, `warn_hi`, `crit_hi`. No vocabulary is consulted; the match is a string operation. Records are keyed by station and sensor, which makes the check conditional on the extraction having attached a station to its threshold records — and that turns out to be the finding rather than a precondition.

Result: the check succeeds completely on three of five runs and not at all on the other two. On runs three, four and five all 22 of the ground truth’s threshold sensors are recovered and 84 of 88 bound cells sit in the field the annotation assigns them to, the same 84 in each run. On runs one and two the threshold records carry no station value, no record can be keyed against the annotation, and the check recovers nothing: 0 of 22 sensors. Pooled over all five runs the figure is therefore 252/440 bound cells, or 57.3%, and that pooled number describes a bimodal outcome rather than an average one.

This is the sharpest available demonstration of what Section 10.2 claims in the abstract. The five runs differ enough that field-level correspondence is impossible to establish on two of them, and $F1_{\text{content}}$ reports

0.719 on all five, identically, with a standard deviation of zero. A difference that destroys slot-level checkability on 40% of runs is completely invisible to the headline metric, because the values are still present in the record and the metric reads a bag of values. Any deployment that needs slot-level guarantees must therefore verify them separately, as this section does, and must expect that verification to fail on runs the headline number cannot distinguish.

The four cells that do mismatch on the three keyable runs are worth examining, because they are not extraction errors.

The residual disagreements are ground-truth errors. The source table for ST03_LABELLING reads, under the printed header `Sensor | Unit | CRIT_LO | WARN_LO | Nominal | WARN_HI | CRIT_HI`:

```
| SPD | m/s | 0.65 | 0.75 | 0.85 +/- 0.01 | 0.95 | 1.05 | ...
```

The pipeline extracts `crit_lo = 0.65`, `warn_lo = 0.75`, `warn_hi = 0.95`, `crit_hi = 1.05`, the document's values, in the document's columns. The ground truth records 1.05, 1.15, 1.35 and 1.05 for the same four fields. Those numbers appear nowhere in the source, and the annotation assigns `critLo` and `critHi` the same value (1.05), which is not a well-formed threshold range. The same pattern holds for ST04_PACKAGING, where the source states `CRIT_LO = 0.85` and the annotation records 1.25. Scored against the source documents rather than the annotation, field placement on the three keyable runs is 88/88 in each — every bound cell the extraction produced carries the value the source document prints, in the column the source document prints it in.

Two conclusions follow, and both bear on how the headline figures should be read. First, where the correspondence can be established at all, the extraction places values correctly at a rate the row-level metric cannot express: a corpus-level $F1_{\text{content}}$ of 0.719 coexists with exact field placement against the source documents on the rules that carry numeric bounds. The two measure different things, and the lower number is not evidence of misplaced values — though neither, as the bimodal result above shows, is the higher one evidence that placement is reliable run to run. Second, the ground truth against which every figure in this thesis is computed contains errors of its own: four specific cells, in a document any reader can open. Reported precision is therefore a lower bound on correctness rather than a measurement of it (Section 10.5).

One practical caveat for any consumer of these records. The column holding the station identifier is not stable across runs: in one run it is `station`, in the others the value sits in `asset` while an empty `station` column also exists. Resolution must therefore take the first non-empty value among candidate columns *per row*, not choose one column per file. This is the naming drift discussed in Section 8.3, appearing exactly where the source document does not print an explicit header for the concept.

9.2.1 A Weaker Binding Check That Does Reach the External Corpora

The verification above requires a column-name correspondence and therefore stops at the development corpus. A weaker form of the same question can be asked of every corpus without constructing any correspondence, using only two things neither side must be told: which of the ground truth's *own* columns holds each role (its low bound is whatever its low-bound column contains), and the internal structure of the predicted record. For every pair the assignment accepted, each populated ground-truth role value is located inside the

paired record: in exactly one scalar field, inside a free-text field, in several fields, or nowhere. If the extractor binds values to slots, the same discovered field should receive a given role in every pair, corpus-wide.

It does. On biogas, the ground truth’s low bound lands in the record field the document calls `h1` in 10 of 10; on desalination the min and max bounds land in one consistent field in 20 of 20 cases each; on the development corpus the modal consistency per threshold role is 0.95–1.00. The sulfuric acid annotation poses no slot-binding question at all: it records every quantity in a single `numeric_value` column, so there are no distinct numeric roles that could be swapped, and the check degenerates. What can be asked there is the weaker record-level question, and the ground-truth value appears within the text of the *matched* record in 121 of 136 checks. Across the three external corpora, 89–100% of ground-truth role values are present in the record the metric paired them with. The check is deliberately weaker than the development-corpus verification, and the weakness has a precise form worth naming: consistency is not correctness. An extractor that placed every low bound in the high slot would do so consistently, and would score a perfect modal share on this test. Consistency establishes that values are bound to slots rather than scattered; it does not establish that the slot is the right one.

That stronger question can be asked wherever the two vocabularies happen to coincide without being made to, which they do on two corpora: the development corpus (`critLo`, `warnLo`, `warnHi`, `critHi` against the pipeline’s `crit_lo`, `warn_lo`, `warn_hi`, `crit_hi`) and desalination (`min_bound`, `max_bound`). For every accepted pair, each ground-truth numeric value is first sought anywhere in the paired record, and only then asked whether it sits under the matching name. The two questions are kept apart deliberately, because a value the extraction never produced is a recall miss and a value it produced but filed elsewhere is a binding error, and adding them together would report the first as the second.

On the development corpus 600 of 920 ground-truth numeric cells are present somewhere in their paired record, and of those 420 — 0.70 — sit in the correspondingly named slot. On desalination all 50 are present and 40 — 0.80 — are correctly placed. These are the only direct measurements of slot correctness this thesis can make, and they should be read as the honest ceiling on what the extraction offers a downstream consumer: roughly three quarters of the values it recovers are where a consumer would look for them, and the remaining quarter are somewhere else in the record. That figure is invisible to $F1_{\text{content}}$ by construction, which is why it is measured here rather than inferred from a headline score.

9.3 The Cost of Requiring No Schema

The grid of Section 7 supplies the prompt with the development corpus’s schema, rule taxonomy, identifier convention, and worked examples drawn from that same benchmark family. Table 8 reports the leading grid conditions rescored with the metric of Section 6.5, so that both sides of the comparison are produced by the same evaluator on the same ground truth. A figure produced by a different metric could not be placed beside these.

Table 8: All three approaches on the development corpus, five repeats each, scored by $F1_{\text{content}}$. Rows are grouped by *what each system was given*, not by score, because that is the variable the table actually manipulates. *Ceiling* is the highest $F1_{\text{content}}$ each system could reach at its own record count (Equation 5) if every record it emitted were correct; *head* is the distance left to it. Only the bottom group can be run on a corpus nobody has configured it for.

System	Given	Recs	$F1_{\text{content}}$	P	R	Ceil.	Head
<i>Per-corpus configuration supplied by a human</i>							
gemma4:31b, reflexion_guided	schema	78.8	0.949 ± 0.020	0.993	0.909	0.956	0.007
gemma4:31b, reflexion	schema	77.2	0.939 ± 0.008	0.993	0.891	0.946	0.007
gemma4:31b, cot_basic	schema	70.4	0.892 ± 0.005	0.992	0.812	0.900	0.008
Hand-written regex parser	layouts + ABox	54.0	0.771	1.000	0.628	0.771	0.000
<i>Nothing supplied</i>							
Multi-agent pipeline	—	131.0	0.719 ± 0.000	0.595	0.907	0.793	0.074
Single schema-free prompt	—	134.0	0.733 ± 0.015	0.601	0.937	0.782	0.049

On F1 alone the table ranks both schema-agnostic conditions last, behind a single schema-specified prompt and a hand-written regex parser. The grouping is what makes that ordering interpretable: every system above the rule was handed something a human produced by reading these four documents, and the two below it were not, so the column the table varies is configuration rather than capability. The two schema-free conditions differ from each other only in orchestration, and are separated by less than either one’s distance to the configured group; Section 9.4 takes that comparison up directly.

Two comparisons are therefore available, and they answer different questions. *Within* the configured group the comparison is like-for-like, and the result is that schema-specified prompting beats hand-written parsing by 0.178 (0.949 against 0.771) — when a human is going to encode the corpus either way, encoding it as a prompt recovers 24 more of the 86 rules than encoding it as regular expressions. *Across* the rule the comparison is not like-for-like and should not be read as one. Four observations develop what it does and does not measure.

Recall is equal. The pipeline recovers 0.907 of the ground truth against the best schema-specified condition’s 0.909: 78 of 86 rows on every repeat in the first case, a mean of 78.2 in the second. The two systems find the same number of rules, and the entire F1 gap opens elsewhere. Whatever the pipeline mainly lacks, it is not coverage of the ground truth.

The entire gap is precision, and the precision gap is granularity. The grid conditions are given the shape of a row, emit 70–79 rows against a ground truth of 86, and carry a precision ceiling of 1.0. The pipeline extracts one record per atomic fact, emits 131 records on every run, and is capped by Equation 5 at 0.656 before any content is examined. The comparison is therefore between a system given the annotator’s granularity convention and one that had to infer it.

The regex parser is not beaten on F1, and no such claim is made. Its 0.771 sits above the pipeline’s 0.719 ± 0.000 , so on this corpus the parser reports the higher F1 and that is stated plainly rather than explained away. What separates them is what each one is. The parser’s precision is perfect for a reason that

is worth stating exactly: its author read these documents and encoded their layouts by hand, and the parser additionally references *only entities the seed knowledge graph declares*, which it reads from that graph’s node file — four stations, their twelve sensors, and seven zones. A parser that can emit no name outside a declared twelve-sensor inventory cannot mis-name an entity, so its 1.000 is a property of what it was given rather than of what it read. This is per-corpus configuration in its purest form, and it is given more of it than either LLM condition: the schema-specified grid is told the shape of a rule but must still recover the entity names from the text, and the pipeline is told neither. The cost is the recall of 0.628: every rule outside the encoded layouts is invisible to it, silently and unrecoverably.

Two structural facts finish the comparison, and both matter more than the ordering. The first is that the parser is *saturated*. Because every record it emits is correct, its 0.771 is exactly the ceiling of Equation 5 for a system emitting 54 records against 86 rows: its headroom is 0.000. It is not a system that is currently performing at 0.771 and might perform better; it is a system that cannot improve by any amount without a human writing more parsing code. The pipeline, at 0.719 against a ceiling of 0.793 at its own record count, retains headroom that belongs to granularity rather than to error.

The second is what “the parser” actually is. It is not one parser applied to four documents but four hand-written functions, one per document, dispatched on the filename — `parse_sop001` through `parse_sop004`, some 480 lines for four files. A fifth document produces nothing at all until somebody writes a fifth function. The unit of effort is therefore the document, not the corpus and not the domain, which is the property that decides where each approach belongs: for a fixed and small set of documents that will not change, hand-written parsing is the correct engineering choice and this thesis does not argue otherwise. The pipeline’s case rests entirely on that condition failing. The pipeline, given none of that, recovers 0.907 of the ground truth — half again as much as the parser, on a corpus the parser’s author had read and the pipeline had not been configured for. The comparison the parser supports is therefore not “LLM pipeline versus simple baseline” but a measurement of what coverage costs when the extraction patterns must be discovered rather than authored; and unlike the pipeline, the parser transfers to a new corpus only by being rewritten.

The schema-specified condition is not a transferable baseline. Its prompt hardcodes field names, class labels, an identifier format, and worked examples containing real station identifiers from the development corpus. Applying it to the biogas, desalination, or sulfuric acid corpora requires a human to read those documents and rewrite the prompt, which is the per-domain configuration this work exists to eliminate, and, more sharply, is an instance of adapting the extractor to the corpus it is scored on.¹

The summary is therefore this. Where a human has already read the documents, that reading is worth roughly 0.23 F1 if it is written into a prompt and 0.05 if it is written as regular expressions, and the prompt is the better use of the same human effort by 0.178. Where nobody has read them, neither configured condition has a score at all — the grid prompt would have to be rewritten and the parser would have to be extended with a function per document — and the pipeline reports 0.649–0.850. The question the table settles is not

¹The claim is about comparability, not mechanical executability, and the distinction was checked rather than assumed: the run registry in the accompanying repository records an exploratory application of the unmodified grid prompts to the desalination document. They emitted 7–14 records against 26 ground-truth rows, expressed in the development corpus’s vocabulary; the outputs were not retained and no figure in this thesis derives from them. The unmodified condition transfers as an under-extraction in a foreign schema, not as a baseline.

which system is best, but what the configuration those systems require is worth, and what it costs when it is unavailable.

9.4 What the Orchestration Buys: A Single-Prompt Ablation

Neither condition in Table 8 isolates the architecture. Both are told things about the corpus that the pipeline is not, so both measure the value of *configuration*. The question left unanswered is narrower and more uncomfortable: does the orchestration — segmentation with line numbering, parallel scouts, the corpus-level inducer, deterministic assembly, the grounding audit — do anything that one call per document would not?

The ablation holds everything constant except the graph. The same model, endpoint, temperature, seed, retry policy and output budget; the same JSON contract and parser; the same evaluator, ground truths and five repeats. The prompt is the scout’s own prompt with only the passages that depend on chunking removed: the record-boundary rules, the attribute-versus-case column test, the field-naming discipline, the verbatim-value rule, the category request and the explicit refusal to supply a worked example are all carried over word for word, because rewriting them would have compared two prompts rather than two architectures. What is removed is the pipeline: no chunking, no line numbers, no inducer, no assembly, no auditor. One document, one call, one row per returned object.

Table 9: The multi-agent pipeline against a single schema-free prompt, five repeats each, same evaluator. Both conditions are given nothing about the target schema; they differ only in orchestration. *Calls* is LLM requests per run (the pipeline issues one scout and one audit per chunk plus one inducer; the baseline issues one per document).

Corpus	Multi-agent		Single prompt		Δ	Fields
	<i>F1</i>	Calls	<i>F1</i>	Calls		
Production line (dev)	0.719 ± 0.000	13	0.733 ± 0.015	4	+0.014	34–44 / 47–55
Biogas digestion	0.715 ± 0.023	7	0.762 ± 0.007	1	+0.047	
RO desalination	0.850 ± 0.019	7	0.793 ± 0.016	1	−0.057	
Sulfuric acid	0.649 ± 0.049	9	0.607 ± 0.014	3	−0.042	
Mean	0.733		0.724		−0.010	

The model is capable of schema-agnostic extraction on its own. That is the first and most useful thing the ablation establishes. Given the same extraction discipline and no target schema, a single call per document reaches $F1_{\text{content}}$ indistinguishable from the full pipeline’s: the two win on two corpora each, and the mean difference of 0.010 sits far inside the run-to-run variation of the corpora themselves, which reaches 0.049. Two individual gaps do exceed both conditions’ spreads and are real — biogas favours the single prompt by 0.047, desalination favours the pipeline by 0.057 — but they point in opposite directions, so no aggregate advantage survives. Complex orchestration is therefore not a precondition for schema-agnostic extraction at this scale, and the result is reported plainly because it is the more useful finding for anyone building such a system.

Two hypotheses about where orchestration should help were tested and not supported. The first is that it buys stability rather than accuracy: it does not, since the single prompt has the *lower* standard deviation on three of the four corpora, including 0.014 against 0.049 on sulfuric acid. Only on the development corpus is the pipeline the steadier, and there its zero variance is a property of the score rather than of the output (Section 9.1). The second is that it should win where a corpus spans several documents, cross-document vocabulary reconciliation being the one thing a per-document call cannot do: it does not, since the pipeline wins on sulfuric acid, which has three documents, and loses on the development corpus, which has four. No property of the corpora tested here predicts the direction of the difference.

What separates the two conditions is not in the score, because the metric does not look there. $F1_{\text{content}}$ evaluates *what* was extracted, never *how* it was obtained, and the two conditions differ almost entirely in the latter. Two differences are concrete and checkable.

The first is provenance. Every pipeline record cites the document and the numbered line it was read from; every baseline record cites nothing, because line numbering was among the stages removed. The consequence is not a matter of degree: the grounding check of Section 10.5 — which establishes that all 669 values asserted by the surplus records appear in the documents they cite — cannot be run on the baseline’s output at all, since there is no citation to check against. An extraction that cannot be traced to a line cannot be audited, and in the settings this work targets that distinction has consequences: an operator configuring an alarm from an extracted threshold needs to see the line of the procedure it came from, and a reviewer disputing a rule needs somewhere to look. The second is vocabulary: the inducer reconciles the development corpus onto 34–44 canonical field names where the baseline’s unreconciled output carries 47–55 for the same documents. A metric that discards field names by construction cannot charge for that sprawl, and this one does not.

The trade-off, stated as measured. The pipeline issues three to seven times as many requests for statistically indistinguishable accuracy: 13 calls against 4 on the development corpus, and 7 against 1 on each single-document external corpus. What those requests buy is not a higher score but line-level traceability and a consolidated vocabulary. A reader whose requirement is a scored rule table and nothing else should prefer the single prompt on this evidence. A reader who must be able to say where each extracted value came from has no such option, because the baseline does not produce that information at any price.

One honest narrowing of that claim. Provenance is delivered by the line-numbering of Stage 1 specifically, not by the graph as a whole. A single prompt given a line-numbered document and asked to cite lines would plausibly recover much of it, and that intermediate condition is not tested here. What the remaining stages contribute — the inducer’s vocabulary reconciliation, the assembly’s record-boundary discipline, the audit’s grounding pass — is therefore separable from the provenance argument and is not individually measured. The trade-off this subsection reports is real and its two halves are both quantified, but attributing all of it to the multi-agent design would credit the architecture for a property one of its stages supplies.

What this settles. It does not show that multi-agent extraction is unnecessary in general; four small synthetic corpora, one model and one prompt family cannot support that. What it settles is that the accuracy in Table 7 is attributable to withholding the schema and to the extraction discipline expressed in the prompt rather than to the graph, and that the graph’s contribution lies in auditability rather than in $F1_{\text{content}}$. Whether

that contribution is worth its cost depends on what consumes the output, which Section 10.13 does not evaluate and does not claim.

10. LIMITATIONS

The limitations below are ordered by how much they bound the claims made in Section 9, most consequential first. Several are properties of the evaluation rather than of the pipeline, and are stated here because a reader cannot otherwise tell how far the reported figures reach.

10.1 Granularity Disagreement Bounds Every Reported Figure

The four corpora disagree about what constitutes one rule, and no convention among them is privileged. Where a ground truth annotates more coarsely than the pipeline extracts, correct records become false positives by the one-to-one assignment (Equation 5); where it annotates more finely (as the development corpus does for threshold rules, splitting one printed table row into up to three annotated rows) ground-truth rows go unmatched and become false negatives instead. Both occur in the same corpus simultaneously (Section 6.5.8), so the discrepancy cannot be removed by a single merge or split rule. On the development corpus the cap is 0.656 on every repeat and the observed precision of 0.595 sits just beneath it. The effect is not a marginal one: on the sulfuric acid corpus, undoing a single annotation convention while leaving the predictions untouched moves the reported figure by 0.147 (Section 6.5.8), which is larger than the gap between several of the corpora in Table 7.

No claim is made that the pipeline’s itemisation is the correct one. Splitting a threshold row into an operational band and a critical trip, or keeping them together, are both defensible schema designs, and which is right depends on what the rows will be used for. The point is only that the two designs are incompatible under a bijective assignment, and that the resulting penalty is charged to whichever side emits more rows regardless of content.

The consequence is that $F1_{\text{content}}$ as reported here conflates two distinct things: whether the pipeline found the right rules, and whether it itemised them the way a particular annotator did. Recall separates them partially (it is unaffected by surplus records) which is why the equal recall between the pipeline and the schema-specified baseline (0.907 against 0.909, Section 9.3) is the more informative comparison. A metric that scored a set of atomic facts against a set of composite rows without requiring a bijection would measure the first thing alone; constructing one is left to future work.

10.2 The Metric Does Not Read Word Order or Attribute Binding

Section 6.6 establishes what $F1_{\text{content}}$ measures, and the same study establishes what it does not. Scrambling token order within a record, against an order-preserving control, costs 0.000; the comparison behaves as a bag of tokens and numbers rather than as a reading of a sentence. Moving quantities between records (the right rule carrying the wrong bound) costs -0.001 – the optimal assignment simply re-pairs each record with the row its new content resembles, absorbing the damage by design, and on the development corpus

the score actually *rises* by 0.017 under the perturbation – and permuting the category label costs +0.007, so neither value-to-rule binding nor label correctness is meaningfully checked.

Two consequences follow. A record that contains the right words and numbers in a nonsensical arrangement is scored almost as highly as a coherent one. And category-label correctness is essentially unpriced, which is consistent with the design argument that category exact-match is unwinnable across vocabularies, but means the pipeline’s category discovery is not evaluated by the headline figure at all. What the metric leaves unchecked about binding is examined by separate means: directly against the ground truth’s fields on the development corpus (Section 9.2) and by the slot-consistency check on all four corpora (Section 9.2.1).

A further, related gap is that the acceptance boundary itself is not calibrated against human judgment: no study here measures whether $\tau = 0.6$ separates pairs a person would call the same rule from pairs they would not. The perturbation results show the metric responds to damage, which is a different property. The audit trail was built so that this check is executable rather than hypothetical (a stratified, blinded sample of assigned pairs, with both blobs and no scores, is included in the accompanying repository as an annotation instrument), but no human labels have been collected, and the threshold’s agreement with human judgment remains unmeasured.

10.3 The Metric Has a Non-Zero Floor

Predictions scored against a foreign corpus’s ground truth do not fall to zero. Pooled per corpus the null score is 0.014–0.037, and the worst pairing reaches 0.101, development-corpus predictions against the biogas ground truth. Reported figures should therefore be read against a floor of roughly 0.10 rather than against 0, and a system scoring below that floor would not be distinguishable from one extracting the wrong document entirely.

The floor is a property of the ground truth being scored against, not a constant, so the usable interval differs per corpus and can be stated exactly. Taking each ground truth’s own floor (the highest score any foreign corpus’s predictions reach against it: 0.101 for biogas, 0.059 for the development corpus, 0.050 for sulfuric acid, 0.027 for desalination) and rescaling the reported figure onto [floor, 1] gives 0.683, 0.701, 0.631 and 0.846 respectively, against the raw 0.715, 0.719, 0.649 and 0.850. Every one of those adjustments is *downward*, by between 0.004 and 0.032: the rescaling subtracts the credit a wrong extraction would receive for free, so by construction it can only lower a reported figure, never raise one. It moves biogas most, because that annotation is the one a foreign extraction scores highest against. It is a reading aid rather than a second metric, and no figure elsewhere in this thesis is rescaled; the headline numbers are the raw ones throughout. Its practical consequence is that scores must be compared within a corpus and not across corpora whose ground truths differ in size.

10.4 The Reported Figures Move With the Acceptance Threshold

The acceptance threshold $\tau = 0.6$ is not a local maximum on any of the four corpora, so it cannot have been chosen to flatter the system, but it is still a choice the figures depend on. All four curves decline through the operating point, at between 0.015 and 0.023 per 0.05 step. At $\tau = 0.4$ the four report 0.793, 0.747, 0.917 and 0.810 against the 0.719, 0.715, 0.850 and 0.649 reported at $\tau = 0.6$, and at $\tau = 0.75$ the development

corpus reports 0.627. Every figure should be read as one point on a curve; all four collapse above $\tau = 0.8$, where almost no pair clears acceptance. On no corpus is the operating point a local maximum, which is the specific form of threshold-fitting the sweep exists to exclude, and on every corpus the reported operating point is the more conservative of the choices available.

Two further consequences follow. The ranking of corpora by $F1_{\text{content}}$ is not stable across the sweep, so no claim that one corpus is intrinsically harder than another is supported. And a single threshold is applied to corpora whose curves have different slopes, which means the four figures in Table 7 are not equally contingent: desalination moves half again as fast through the operating point as the development corpus does.

A related dependence is the tag-derived numbers of Section 6.5.5. Excluding them leaves the development corpus unmoved, raises biogas by 0.010 and desalination by 0.067, and lowers sulfuric acid by 0.048. The reported figures are those of the declared default in every case; the point of measuring the toggle is that its effect is neither uniform nor uniformly favourable, and a reader can see which corpora it reaches.

10.5 Ground-Truth Incompleteness

Two distinct annotation problems affect the reported figures, and they push in the same direction.

Omission. The development corpus’s unpaired records state facts the source documents contain but the annotation schema has no row for: inter-station dependencies recorded in a table the ground truth does not cover, or the `nominal` column of the threshold tables, which the documents print and the annotation drops. These are counted as false positives.

That claim is checked rather than asserted, because the alternative explanation — that the surplus is invention — would undermine every granularity argument in this thesis. Each record cites the document and line it was read from, so the check is deterministic and needs no model: take the 225 unpaired records across the five development runs, and ask whether the field values they assert appear in the source document they cite. *All 669 values do.* No unpaired record asserts a value absent from its own document, and the eight that carry numeric values have all 26 of those values present on the specific line cited.

Two qualifications keep that figure honest. First, at the stricter scope of the individual cited line rather than the whole document, 459 of 669 values (68.6%) are quotable; the shortfall is almost entirely matrix records, whose subject is a *column header* sitting on a different line from the cell they cite — including the four headers the parser corrupts (Section 10.7), which the records copy faithfully and which therefore match nothing spelled correctly. Second, a substring test against a whole document could be vacuous, so it is given the same null case as the metric: scored against a document they did not come from, the same values match at 20.9% rather than 100%. The test discriminates, and the grounded fraction should be read against that floor.

What this establishes is bounded and worth stating precisely. It shows the surplus records are not fabrications, which is what the granularity argument requires. It does not show they are *useful*, nor that the annotator was wrong to omit them, and Section 6.5.3 has already conceded the downstream cost of emitting them.

Error. The ground truth is also, in places, simply wrong. Four bound cells for the two SPD sensors

record values that appear nowhere in the source documents, and for ST03_LABELLING they assign `critLo` and `critHi` the same value, which is not a well-formed range. The pipeline extracts the printed values correctly and is scored as incorrect for doing so (Section 9.2). These are counted as false positives. The reported precision is therefore a lower bound on the fraction of records that are *correct*, and an accurate measure only of the fraction that are *annotated*. Quantifying the difference would require re-annotating the corpora at the pipeline’s granularity, which would compromise their independence.

10.6 Results Are Not Bit-Reproducible

Every agent role is served by a hosted endpoint whose long generations are not reproducible under a fixed seed, for the reasons given in Section 6.3.3. Repeats therefore differ, and every figure is a mean over five runs with its standard deviation. Exact replay is available only through the response cache, which is keyed on the request content and is disabled whenever repeats are requested, so the reported spreads are real, but no single number in this thesis can be reproduced exactly without that cache. The sulfuric acid corpus’s standard deviation of 0.049, spanning 0.610–0.733, shows the spread is not always negligible, and a corpus reporting a standard deviation of zero is not thereby deterministic: on the development corpus the five runs differ from one another in field naming and category labelling while converging on the same score, because those are precisely the differences $F1_{\text{content}}$ is built not to charge for (Section 9.1). Five repeats therefore bound neither the spread of the endpoint nor the spread of the underlying outputs. The irreproducibility is in the extraction and not in the scoring: the evaluator is deterministic, so rescoring the retained predictions reproduces every figure in this thesis exactly, and the accompanying repository contains the predictions that make that check possible.

10.7 The Document Parser Is Unmeasured and Barely Exercised

Section 5 describes the layout-aware conversion that turns a source PDF into the numbered text every later stage reads. No figure in this thesis measures how faithful that conversion is, and two facts make the omission worth stating rather than leaving implicit.

The first is coverage. Only the development corpus passes through the parser; the three external corpora are plain text as authored, so every transfer result reported here exercises the extraction pipeline but not the component that feeds it. Whatever the parser contributes to or subtracts from the development figures, it contributes nothing to the external ones, and the two are therefore not produced by identical toolchains.

The second is that the conversion demonstrably introduces errors. The access matrix in the development corpus’s fourth document has seven zone columns, and four of their headers are corrupted in the parsed text: a space is inserted inside the last word of three of them (Gen. Warehous e, Main Entranc e, Chem. Stora ge) and a spurious semicolon appears in the fourth (R&D; Lab). These are the column labels of the matrix whose cells state the 42 role–zone authorisations discussed in Section 10.5, so the damage falls on the subject of a substantial block of rules. The extraction copies them faithfully, which is the behaviour the scout prompt demands and the only behaviour that keeps provenance honest, and is then compared against an annotation that spells the same zones correctly. The resulting mismatch is charged to the extractor.

No claim is made here about the size of that effect, because it was not measured; the point is that the reported figures for the development corpus include an unquantified contribution from a preprocessing step, that no external figure includes it at all, and that improving extraction accuracy on scanned or PDF-native documents may be a question about the parser rather than about the pipeline. Establishing parser fidelity against a hand-transcribed reference is the obvious missing measurement and is left to future work.

10.8 Corpus Scale and Composition

The four corpora comprise nine documents and 229 annotated rules in total, with two of the three external corpora consisting of a single document each. This is enough to show that performance does not collapse on unseen material, and not enough to characterise how performance varies with document length, corpus size, or domain distance. All documents are synthetic and English; real industrial SOPs would introduce annotation noise, inconsistent formatting, and multilingual content that this evaluation does not probe. One further composition limit should be stated plainly: the documents and their annotations were authored by the same process, so the annotations agree with their documents about what a rule is far more closely than an independently annotated corpus would. The sulfuric acid corpus is the one case where document and annotation were produced without the pipeline’s conventions in view, and it is also the lowest-scoring of the four — which is the direction that observation would predict.

10.9 Five Repeats Bound the Statistical Claims

Five runs per configuration supports reporting a mean and a standard deviation and supports declaring two overlapping intervals tied. It does not support significance testing, and no such test is claimed. Where two figures differ by less than the standard deviations separating them, this thesis reports them as indistinguishable rather than ranking them.

10.10 The External Corpora Were Not Equally Held Out

“Held out” admits degrees, and describing all three external corpora with the same phrase would overstate two of them. No external corpus ever contributed to a configuration choice: every prompt, parameter, model and threshold in this thesis was selected against the development corpus, and the selection is separable from the evaluation for the structural reason given in Section 4. That is a weaker guarantee than never having been looked at, and the difference matters.

Biogas and desalination were scored at several points while the pipeline was being developed. No number they produced was used to tune a parameter, but their scores were visible to the author between revisions, and a component reworked after seeing a disappointing external figure has been influenced by that figure whether or not any constant was edited. This is adaptive exposure rather than tuning, and it cannot be undone retrospectively; it can only be declared. The sulfuric acid corpus is the one case free of it. It was constructed after the prompts, parameters and evaluator had been frozen, scored once with five repeats, and reported — and it returns the lowest figure of the four, 0.649, which is the direction adaptive exposure predicts for the corpora that had it.

A second property of the material must be declared alongside this. Every document in this evaluation is synthetic. The development corpus comes from a published synthetic benchmark; the sulfuric acid corpus and its annotation were both generated by a language model from a specification that fixed the container format — documents as plain text, annotation as one row per rule — and fixed nothing about the schema, the layout, or the vocabulary, which the generator chose for itself. That construction is what allowed a genuinely unseen corpus to be added after freezing, and it is also its principal weakness: a generated document is more internally consistent than a real one, and a generated annotation agrees with its document about what a rule is more readily than an independent human annotator would. Neither the documents nor the annotation were inspected against the pipeline’s behaviour before being scored.

What the four corpora do establish should be stated as precisely as what they do not, because the distinction is the difference between a weak benchmark and a narrow one. The variable this thesis manipulates is *structural heterogeneity* — whether rules arrive as prose clauses, as attribute tables, as printed-identifier blocks, or as a mixture — and on that variable the four corpora genuinely differ, which is why they produce four different failure modes rather than four instances of one. A synthetic document can vary its layout as freely as a real one, and the pipeline’s indifference to layout is therefore tested here in the way it is claimed. What a synthetic corpus cannot supply is the disagreement between a document and an annotator who did not write it: the transcription noise, the inconsistent conventions, and the genuine ambiguity about what constitutes one rule that a real facility’s paperwork carries. Those are precisely the conditions under which the granularity effect of Section 6.5.8 would be largest, so the figures here should be read as an upper bound with respect to annotation agreement, and no claim about real deployment should be read out of them.

10.11 The Figures Move With the Prompts’ Illustrative Vocabulary

No stage is given a target schema, a rule taxonomy, or a worked example, but the prompts are not vocabulary-free: they must illustrate what a printed rule identifier looks like, what kinds of subject a table’s axes might range over, and what a functional category is, and every such illustration is a word a human chose. Those illustrations are drawn from outside the evaluated material, and the constraint is enforced mechanically rather than by inspection — no example token may appear as a printed identifier in an evaluated document, as a column name of any annotation, or as an annotation’s category or severity label.

The choice is not cost-free, and its cost is measured rather than asserted. A variant of the prompts whose illustrations instead share vocabulary with the corpora’s annotations — example identifier codes of the shape those documents print, and kind-of-subject nouns matching those annotations’ column names — scores *higher* on two of the four corpora: +0.081 on biogas and +0.037 on the development corpus, against +0.007 on desalination, which is inside its run-to-run spread. On biogas the mechanism is visible in the outputs rather than merely inferred: the variant emits 36.4 records against the compliant prompts’ 28.0, itemising at a granularity closer to that annotation’s own instead of bundling several rules into one record.²

Three things follow, and the last is the uncomfortable one. The first is a distinction the result should not

²These four figures are the one group in this thesis still computed under the earlier, recall-only form of the numeric term (Section 6.5.5). The variant’s predictions were not retained, so they cannot be rescored without re-running extraction, which would confound the comparison with the endpoint’s own run-to-run variation. They are reported as measured and should be read as indicative of the direction and rough size of the effect rather than as directly comparable with the figures in Table 7.

be allowed to blur: schema-agnostic is a claim about what the pipeline is *given* — no target schema, no rule taxonomy, no worked example, and no token appearing as a field name or category label in any evaluated annotation — and that claim is enforced mechanically and remains exactly true. It is not a claim that the pipeline is *insensitive* to how its instructions are worded, and this measurement shows it is not. Agnostic and insensitive are different properties; only the first is claimed, and conflating them would overstate the result in one direction as surely as ignoring the 0.081 would in the other. Second, reported figures are a lower bound with respect to this degree of freedom, since the alternative was measured and is higher. But it also means a reported figure on this kind of task is partly a function of how well an illustration in a prompt happens to match an annotator’s vocabulary — a quantity no metric in this thesis observes, that a reader cannot see from a results table, and that is easily supplied without any intent to do so. The comparison is a package difference rather than a controlled ablation: the two prompt variants differ at roughly eight sites simultaneously, so the 0.081 cannot be attributed to any one substitution, and part of it may be ordinary wording quality rather than vocabulary overlap. Separating them would require varying one illustration at a time, which is left to future work.

10.12 Single Model, Single Endpoint

All three agent roles are served by one model, `gemma4:31b`, selected on grid evidence over one alternative. Whether the architecture’s behaviour generalises across model families is untested, and the specific finding that role independence between scout and auditor is not worth its cost rests on a comparison with a single weaker alternative rather than on a survey.

10.13 No Downstream Validation

This thesis evaluates rule extraction against annotated ground truth. It does not demonstrate that the extracted rules are sufficient to drive a downstream application such as knowledge-graph population or sensor-stream anomaly detection. A high $F1_{\text{content}}$ establishes agreement with an annotator, which is a necessary but not a sufficient condition for operational usefulness; the two can diverge, for instance where the metric’s insensitivity to attribute binding (Section 10.2) admits a record whose bound is attached to the wrong sensor. The accompanying repository contains preliminary downstream tooling of exactly this kind (knowledge-graph population and sensor-stream anomaly detection over extracted rules, under `layers/layer_4`); it is not integrated with the pipeline evaluated here and none of its output is reported in this thesis. Establishing operational sufficiency requires closing that loop and is left to future work.

11. CONCLUSION

This thesis asked what rule extraction from industrial operating documents costs when nothing about the target schema is supplied. The question is worth asking because the alternative is well understood and does not scale: an extractor given a facility’s field names, rule taxonomy, identifier convention and worked examples performs very well on that facility and cannot be moved to the next one without a human reading

its documents and rewriting the prompt. The measurement reported here puts a number on that trade rather than assuming its direction.

11.1 What Was Built and What Was Measured

The pipeline segments each document and numbers every content line in code, so that provenance is something the system knows rather than something a model must remember to report. Parallel scouts read the numbered chunks and name fields from the document’s own labels; a single inducer call reconciles those labels into one corpus-level schema while never seeing a document; deterministic assembly emits one row per rule; and parallel auditors re-read each record against the lines it cites. No stage receives a target schema, a rule taxonomy, or a worked example.

Across one development corpus and three external corpora, five repeats each and no per-corpus configuration, it reaches $F1_{\text{content}}$ of 0.719 on the development corpus and 0.715, 0.850 and 0.649 on the three external ones. The development corpus is neither the best nor the worst of the four. That is the central result, and it is deliberately a modest one: it shows that performance does not fall away on material the system was not built against, which is the failure a development corpus exists to expose, and it does not show that unseen corpora are handled uniformly, because they are not.

Against this, a schema-specified grid reaches 0.949 on the development corpus, and a hand-written regex parser 0.771. Neither transfers: the grid’s prompt encodes that corpus’s vocabulary, and the parser encodes its layouts and draws its entity names from the corpus’s own seed knowledge graph. The informative comparison is not the F1 ordering but its decomposition. Recall is near-identical between the pipeline and the best schema-specified condition (0.907 against 0.909, 78 of 86 rows in both cases) while the parser reaches 0.628; the entire gap to the grid is precision, and the precision gap is granularity rather than error.

11.2 The Trade-Off the Architecture Actually Represents

The orchestration was then removed while everything else was held constant — same model, same extraction discipline in the prompt, same evaluator, one call per whole document instead of a graph over chunks. It scores 0.733, 0.762, 0.793 and 0.607, winning on two corpora and losing on two, for a mean difference of 0.010 against run-to-run variation reaching 0.049, at three to seven times fewer requests. Two hypotheses about where orchestration should help were tested and not supported: the single prompt is the *steadier* of the two on three corpora, and the pipeline’s advantage does not appear on the multi-document corpora where cross-document reconciliation should have helped.

The first thing this establishes is that the model is capable of schema-agnostic extraction unaided, which is useful to anyone building such a system and is not obscured here. The second is what the metric cannot see. $F1_{\text{content}}$ scores *what* was extracted, never *how*, and the two conditions differ almost entirely in the latter: every pipeline record cites the document and line it came from, and every baseline record cites nothing. The grounding check of Section 10.5 cannot be run on the baseline’s output at all, for want of anything to check against, and the pipeline additionally reconciles the development corpus onto 34–44 field names where the baseline carries 47–55.

The trade-off is therefore between request count and traceability, and both halves are measured. The claim needs one narrowing to stay exact: provenance is supplied by Stage 1’s line numbering rather than by the graph as a whole, and a single prompt over a line-numbered document might recover much of it — an intermediate condition this thesis does not test. What the remaining stages contribute beyond that is not separately measured, and whether traceability repays its cost depends on a downstream consumer that Section 10.13 does not evaluate.

11.3 What the Evaluation Establishes About Its Own Metric

A schema-agnostic system cannot be scored by a schema-aware metric, so $F1_{\text{content}}$ was built alongside it and then tested rather than trusted. It discriminates: predictions scored against a foreign corpus’s ground truth fall to between 0.01 and 0.04, corrupting every numeric value costs 0.20, and padding a record with numbers it did not extract costs a further 0.10 rather than earning anything — the last of these only because the numeric term scores both directions, a property whose absence in an earlier form of the metric made padding profitable. It is also blind in ways that are stated rather than discovered by a reader — it does not read word order, it does not check which record a quantity is bound to, it prices category labels at almost nothing, and, most consequentially for an industrial setting, reversing every bound inside every record costs it exactly nothing. Attribute binding is therefore verified outside the metric, deterministically. On the one corpus where a field correspondence exists without inventing one, that check returns a bimodal result: on three of five runs every bound cell sits in the column the source document prints it in, and on the other two the threshold records carry no station and cannot be keyed at all — a difference the headline metric, which reports an identical 0.719 on all five, cannot see.

The most consequential finding about the metric is that it cannot see annotation granularity. Regrouping two attribute tables in the sulfuric acid annotation — changing no prediction whatever — moves the reported figure by 0.147. Any one-to-one row-matching metric shares this property. It bounds how finely cross-corpus figures of this kind should be read, in this thesis and elsewhere.

11.4 Contributions

- A rule-extraction pipeline that assumes only that its input is an operating document, whose field vocabulary, record boundary and category set are derived from the corpus rather than supplied, and which runs unchanged across four unrelated industrial domains.
- A schema-agnostic evaluation metric, together with the perturbation, null-pairing and sensitivity studies that establish what it measures and the four things it does not.
- A measurement of what schema specification is worth on a corpus that has one (0.230 F1) and what it costs on a corpus that does not (the condition cannot be run at all without being rewritten).
- An ablation isolating the orchestration from the schema-agnostic prompting it runs, which finds the two indistinguishable in accuracy (0.010 mean F1 over four corpora, two won each way) at three to

seven times the request count, locating the architecture’s contribution in traceability rather than in score.

- A quantification of two degrees of freedom that evaluations of this kind rarely report: annotation granularity, worth 0.147 on identical predictions, and the prompts’ illustrative vocabulary, worth up to 0.081.

11.5 Closing Assessment

The honest summary is that schema-agnostic extraction on these corpora is worth roughly 0.23 F1 less than schema-specified extraction on the one corpus where both can run, that most of that difference is a disagreement about what constitutes one rule rather than about what the documents say, that the resulting system transfers to a new facility at the cost of running it, and that the multi-agent orchestration is not what delivers the accuracy — a single prompt withholding the same schema reaches the same score for a fraction of the requests, and what the orchestration buys instead is the line-level traceability that makes an extracted value auditable. Whether that trade is worth making is a deployment question and depends on how many facilities there are.

Two things would most improve confidence in these numbers, and neither is a larger model. The first is corpora that are real rather than synthetic and annotated by people who did not also write the documents, which would test the granularity agreement this evaluation cannot. The second is a metric that scores a set of atomic facts against a set of composite rows without requiring a bijection; the 0.147 measurement above is a lower bound on what such a metric would be worth, and constructing one would remove the largest source of ambiguity in every figure reported here.

A. COMPLETE EVALUATION RESULTS

This appendix lists every scored condition. All figures are produced by the same evaluator (Section 6.5) with an acceptance threshold of 0.6, so every row is directly comparable to every other. *Records* is the mean number of extracted rows; *Max P* is the ceiling the record count imposes on precision (Equation 5).

Multi-Agent Pipeline (no schema supplied)

Table 10: Schema-agnostic multi-agent pipeline, five repeats per corpus. Development corpus above the rule, external corpora below it, ordered as in Table 2 rather than by score.

Corpus	GT	Records	F1	SD	P	Max P	R
dev_production_line	86	131.0	0.719	0.000	0.595	0.656	0.907
external_test_biogas	47	28.0	0.715	0.023	0.957	1.000	0.570
external_test_desalination	26	27.2	0.850	0.019	0.831	0.956	0.869
external_test_sulfuric_acid	70	49.0	0.649	0.049	0.788	1.000	0.552

Single Schema-Free Prompt (orchestration removed)

One call per whole document, no chunking, no schema induction, no assembly and no audit, with the model, prompt discipline and evaluator held constant (Section 9.4).

Table 11: Single schema-free prompt, five repeats per corpus, scored by $F1_{\text{content}}$ against the same ground truths as Table 10.

Corpus	GT	Records	F1	SD	P	Max P	R
dev_production_line	86	134.0	0.733	0.015	0.601	0.642	0.937
external_test_biogas	47	33.8	0.762	0.007	0.911	1.000	0.656
external_test_desalination	26	30.0	0.793	0.016	0.740	0.867	0.854
external_test_sulfuric_acid	70	48.0	0.607	0.014	0.746	1.000	0.511

Schema-Specified Grid, Rescored with the Current Metric

All conditions below were prompted with the development corpus’s schema, taxonomy, identifier convention and worked examples (Section 7.1), and are scored against that corpus only. They cannot be applied to the external corpora as comparable conditions without their prompts being rewritten.

Table 12: All grid conditions and the regex baseline, rescored by $F1_{\text{content}}$ on the development corpus.

Condition	Runs	Records	F1	P	R
gemma4-31b_reflexion_guided	5	78.8	0.949	0.993	0.909
gemma4-31b_reflexion	5	77.2	0.939	0.993	0.891
gemma4-31b_pre_act	5	70.6	0.894	0.992	0.814
gemma4-31b_cot_basic	5	70.4	0.892	0.992	0.812
gemma4-31b_naive	5	66.2	0.859	0.988	0.760
gemma4-31b_cot_structured	5	63.6	0.850	1.000	0.740
gemma4-31b_self_consistency	5	63.0	0.846	1.000	0.733
gemma4-31b_few_shot_static	5	62.2	0.840	1.000	0.723
nemotron-3-nano-30b_reflexion_guided	5	67.6	0.840	0.957	0.751
gemma4-31b_graph_informed	5	62.0	0.838	1.000	0.721
nemotron-3-nano-30b_self_consistency	5	60.4	0.825	1.000	0.702
nemotron-3-nano-30b_cot_basic	5	62.0	0.822	0.983	0.709
nemotron-3-nano-30b_naive	5	60.4	0.819	0.993	0.698
gemma4-31b_react_abox	5	59.2	0.814	1.000	0.688
nemotron-3-nano-30b_reflexion	5	79.0	0.794	0.871	0.756
baseline_b0	1	54.0	0.771	1.000	0.628
nemotron-3-nano-30b_pre_act	5	97.8	0.768	0.795	0.805
nemotron-3-nano-30b_few_shot_static	5	67.2	0.767	0.906	0.681
nemotron-3-nano-30b_graph_informed	5	57.4	0.754	0.947	0.630
nemotron-3-nano-30b_cot_structured	5	78.4	0.729	0.859	0.693
nemotron-3-nano-30b_react_abox	5	49.8	0.707	0.972	0.563

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